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CHEMISTRY • UPPER SIXTH | UPPER SIXTH — CHEMISTRY

CHAPTER 5

ENVIRONMENTAL CHEMISTRY: SULPHUR AND NITROGEN

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter explores the chemistry of sulphur and nitrogen, two elements with wide ranges of oxidation states that underpin the manufacture of sulphuric acid, ammonia, nitric acid and fertilizers. It also examines how oxides of sulphur and nitrogen pollute the atmosphere, cause acid rain and eutrophication, and how these effects can be controlled.

- By the end of this chapter I can assign oxidation numbers to sulphur (−2, +2, +4, +6) and nitrogen (−3 to +5) in their common compounds and use them to classify redox behaviour.
- By the end of this chapter I can describe the Contact process for manufacturing sulphuric acid, with conditions, equations and the role of the V2O5 catalyst.
- By the end of this chapter I can state and explain the physical and chemical properties of concentrated and dilute sulphuric acid (acid, oxidant, dehydrating agent).
- By the end of this chapter I can describe the Haber process for ammonia synthesis and justify the compromise conditions using Le Chatelier's principle.
- By the end of this chapter I can describe the Ostwald process for nitric acid manufacture from ammonia.
- By the end of this chapter I can write balanced equations for the reactions of ammonia (basicity, reducing agent, complex formation) and of nitric acid (acid and oxidizing agent).

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SUMMARY SHEET the essentials to remember

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1. Sulphur and its Variable Oxidation States
2. Sulphuric Acid: Manufacture, Properties and Uses
3. Oxides of Sulphur and Atmospheric Pollution
4. Nitrogen, Ammonia and Nitric Acid
5. Fertilizers and their Environmental Effects
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can assign oxidation numbers to sulphur (−2, +2, +4, +6) and nitrogen (−3 to +5) in their common compounds and use them to classify redox behaviour.
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- By the end of this chapter I can write balanced equations for the reactions of ammonia (basicity, reducing agent, complex formation) and of nitric acid (acid and oxidizing agent).
- By the end of this chapter I can explain how SO2 and NOx form, how they produce acid rain and photochemical smog, and describe control measures.
- By the end of this chapter I can explain the composition and manufacture of common NPK fertilizers and analyse their environmental effects, especially eutrophication.
- By the end of this chapter I can carry out mole and percentage-yield calculations on industrial processes (Contact, Haber) in GCE-style problems.

PREREQUISITES

- Oxidation numbers and balancing redox equations (Lower Sixth: redox reactions)
- Chemical equilibrium and Le Chatelier's principle (Lower Sixth)
- Rates of reaction and catalysis (Lower Sixth)
- Mole calculations, molar gas volume and percentage yield (Fifth Form / Lower Sixth)
- Acid-base behaviour: acids, bases, salts and pH (Lower Sixth)

1 SULPHUR AND ITS VARIABLE OXIDATION STATES

In Douala's industrial zone and around the SONARA refinery in Limbe, residents sometimes notice a sharp, choking smell after gas flaring — a sign of sulphur dioxide escaping into the air. Meanwhile, farmers in the Western Highlands spread ammonium- and nitrate-based fertilizers to boost maize and potato yields, yet streams downhill sometimes turn green with algae. Both stories involve two elements, sulphur and nitrogen, whose many **oxidation states** make them industrially precious but environmentally dangerous. How are sulphuric acid and ammonia manufactured on a massive scale, and why do their by-products acidify rain and suffocate lakes? This chapter answers these questions by following sulphur and nitrogen from the factory to the field to the atmosphere. We begin with sulphur itself: where it comes from, what forms it takes, how it reacts, and how to keep track of its oxidation numbers.

1.1 Occurrence and Extraction of Sulphur

Sulphur is one of the most abundant elements in the Earth's crust. It occurs in two main forms:

- **Native (elemental) sulphur**, found in underground deposits, often associated with salt domes and volcanic regions;
- **Combined sulphur**, in minerals such as iron pyrites FeS_2, galena PbS, zinc blende ZnS and gypsum $CaSO_4 \cdot 2\,H_2O$, and as hydrogen sulphide H_2S in crude oil and natural gas ("sour gas").

Today, most of the world's sulphur no longer comes from mined deposits but is recovered from **crude oil and natural gas refining**. Natural gas containing H_2S is treated so that the H_2S is separated and then converted to sulphur by the **Claus process**: part of the H_2S is burnt to SO_2, which then reacts with the remaining H_2S over a catalyst:

$$2\,H_2S(g) + SO_2(g) \longrightarrow 3\,S(s) + 2\,H_2O(g)$$

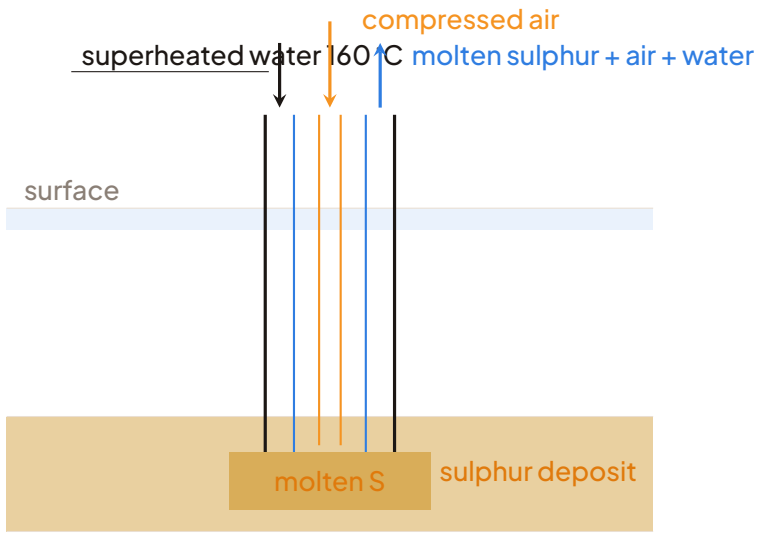
Where deep underground deposits of elemental sulphur exist, they are mined by the elegant **Frasch process**, which exploits sulphur's low melting point (about 115 °C).

EXPERIMENT **The Frasch process**

Three concentric steel pipes are sunk into a borehole reaching the sulphur deposit.

1. Superheated water at about 160 °C (kept liquid under pressure) is pumped down the outer pipe; it melts the sulphur.
2. Compressed air is forced down the innermost pipe.
3. The air mixes with the molten sulphur, making it light and frothy, so that the mixture of molten sulphur, water and air is pushed up the middle pipe to the surface.

At the surface the sulphur solidifies in large vats; it is about 99.5% pure.



The Frasch process: three concentric pipes deliver superheated water and compressed air to melt and lift underground sulphur.

1.2 Allotropes of Sulphur

If you dissolve sulphur in carbon disulphide and let the solvent evaporate slowly, you obtain beautiful yellow crystals. If instead you melt sulphur and let it crystallise just below its melting point, you get needle-shaped crystals of a different structure. Same element, same physical state (solid), but different crystalline forms: this is allotropy.

DEFINITION **Allotropy**

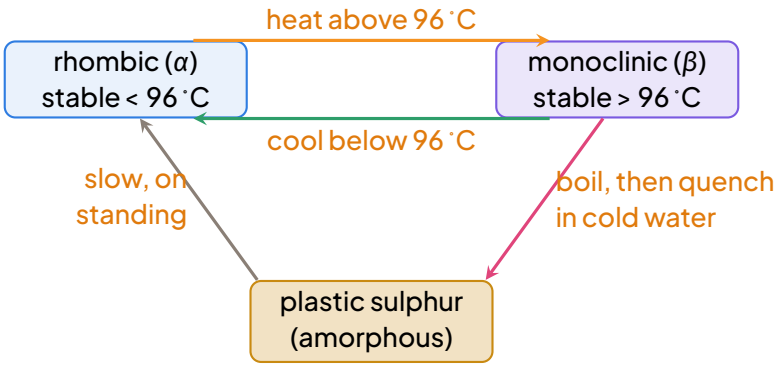
Allotropy is the existence of an element in more than one form in the same physical state. The different forms are called **allotropes**; they differ in the arrangement or bonding of their atoms and therefore in some physical properties.

Sulphur has two crystalline allotropes, both built from puckered S_8 rings packed differently in the crystal:

- **Rhombic (α) sulphur**: stable below 96 °C; octahedral-shaped crystals; density about 2.06 g cm^{-3}.
- **Monoclinic (β) sulphur**: stable above 96 °C up to the melting point; needle-shaped (prismatic) crystals; density about 1.96 g cm^{-3}.

The temperature 96 °C is the **transition temperature**: below it rhombic sulphur is stable, above it monoclinic sulphur is stable. The change is slow, so each form can persist for a while in the other's range.

A third, non-crystalline form is **plastic sulphur**: when boiling sulphur is poured into cold water (quenched), the S_8 rings break into long chains that have no time to crystallise, giving a brown, rubbery, amorphous solid which slowly reverts to rhombic sulphur on standing.



Interconversion of the allotropes of sulphur. Rhombic and monoclinic sulphur interconvert at the transition temperature 96 °C; quenching boiling sulphur gives amorphous plastic sulphur.

1.3 Physical Properties of Sulphur

KEY POINTS **Physical properties of sulphur**

- Yellow, brittle, non-metallic solid with no taste and no smell (the "smell of sulphur" is actually due to compounds such as SO_2 or H_2S).
- Insoluble in water, but soluble in carbon disulphide CS_2 (and in organic solvents such as benzene).
- Low melting point, about 115 °C; poor conductor of heat and electricity.

1.4 Chemical Properties of Sulphur

Burning in air or oxygen. Sulphur burns with a blue flame to give the choking gas sulphur dioxide:

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

Reaction with metals. Sulphur combines directly with many metals on heating to give **sulphides**. A classic school demonstration is the vigorous, glowing reaction of iron filings with sulphur:

$$Fe(s) + S(s) \longrightarrow FeS(s)$$

Note that the product, iron(II) sulphide, is a compound: it can no longer be separated by a magnet, unlike the original mixture.

Reaction with hydrogen. On passing hydrogen over boiling sulphur, hydrogen sulphide forms:

$$H_2(g) + S(l) \longrightarrow H_2S(g)$$

1.5 Assigning Oxidation Numbers of Sulphur

The key to understanding sulphur chemistry is its **variable oxidation state**: sulphur appears at −2, 0, +2, +4 and +6. We track these using oxidation numbers.

METHOD **Assigning oxidation numbers**

1. A free (uncombined) element has oxidation number 0: S, O_2, Fe.
2. Oxygen is usually −2 (except in peroxides, −1, and with fluorine).
3. Hydrogen is usually +1 (except in metal hydrides, −1).
4. The sum of oxidation numbers equals the overall charge of the species (0 for a neutral molecule).

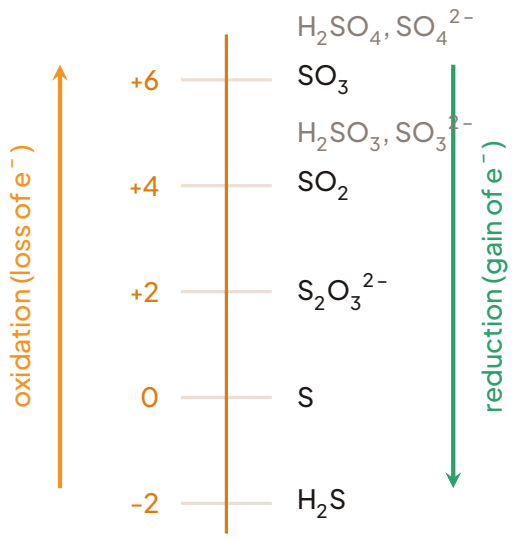
EXEMPLE **Oxidation number of S in H_2SO_4**

Let x be the oxidation number of sulphur. Then

$$2(+1) + x + 4(-2) = 0 \quad \Longrightarrow \quad x = +6.$$

Similarly, in $S_2O_3^{2-}$ (thiosulphate): $2x + 3(-2) = -2$, so $x = +2$ (an average value).

Species	Name	Oxidation number of S
H_2S	hydrogen sulphide	−2
S	elemental sulphur	0
$S_2O_3^{2-}$	thiosulphate ion	+2 (average)
SO_2, H_2SO_3	sulphur dioxide, sulphurous acid	+4
SO_3, H_2SO_4	sulphur trioxide, sulphuric acid	+6



The oxidation-state ladder of sulphur, from -2 to +6, with example species at each rung.

1.6 Hydrogen Sulphide H_2S: Sulphur at -2

Preparation. In the laboratory, H_2S is prepared by the action of dilute hydrochloric acid on iron(II) sulphide, usually in a Kipp's apparatus:

$$FeS(s) + 2\,HCl(aq) \longrightarrow FeCl_2(aq) + H_2S(g)$$

WARNING **H_2S is highly toxic**

Hydrogen sulphide is extremely poisonous — as toxic as hydrogen cyanide — despite its familiar rotten-egg smell. Worse, it deadens the sense of smell at high concentration, so the disappearance of the smell is not a sign of safety. Never prepare or use H_2S outside a fume cupboard.

Properties.

- Colourless gas with a rotten-egg smell; denser than air; fairly soluble in water.
- Its aqueous solution is a **weak dibasic acid**, ionising in two steps:

$$H_2S(aq) \rightleftharpoons H^+(aq) + HS^-(aq) \qquad HS^-(aq) \rightleftharpoons H^+(aq) + S^{2-}(aq)$$

  It forms two series of salts: hydrogensulphides (e.g. NaHS) and sulphides (e.g. Na_2S).
- It is a **strong reducing agent**, because sulphur at -2 can only be oxidised. For example, it decolourises acidified potassium manganate(VII) and reduces iron(III) to iron(II):

$$2\,FeCl_3(aq) + H_2S(g) \longrightarrow 2\,FeCl_2(aq) + 2\,HCl(aq) + S(s)$$

Test for H_2S. The gas turns filter paper moistened with **lead(II) ethanoate** solution black, owing to the formation of lead(II) sulphide:

$$Pb(CH_3COO)_2(aq) + H_2S(g) \longrightarrow PbS(s) + 2\,CH_3COOH(aq)$$

1.7 Sulphur Dioxide SO_2 and Sulphurous Acid H_2SO_3: Sulphur at +4

Preparation. In the laboratory, SO_2 is conveniently prepared by the action of a dilute acid on a sulphite:

$$Na_2SO_3(s) + 2\,HCl(aq) \longrightarrow 2\,NaCl(aq) + H_2O(l) + SO_2(g)$$

Industrially it is obtained by burning sulphur or roasting sulphide ores.

Acidic oxide. SO_2 dissolves in water to form **sulphurous acid**, a weak dibasic acid:

$$SO_2(g) + H_2O(l) \rightleftharpoons H_2SO_3(aq)$$

It turns moist blue litmus paper red and reacts with alkalis to give sulphites SO_3^{2-} and hydrogensulphites HSO_3^-.

Reducing agent (the usual behaviour). With sulphur at +4, SO_2 is readily oxidised to +6, so it is a good **reducing agent**. It decolourises acidified potassium manganate(VII):

$$2\,MnO_4^-(aq) + 5\,SO_2(g) + 2\,H_2O(l) \longrightarrow 2\,Mn^{2+}(aq) + 5\,SO_4^{2-}(aq) + 4\,H^+(aq)$$

(purple → colourless), and turns acidified dichromate(VI) from orange to green.

Oxidizing agent (with stronger reducers). With a stronger reducing agent such as H_2S, SO_2 behaves as an oxidizing agent, being reduced to sulphur (+4 → 0):

$$SO_2(g) + 2\,H_2S(g) \longrightarrow 3\,S(s) + 2\,H_2O(l)$$

PROPERTY **Dual redox character of SO_2**

Sulphur dioxide is both a reducing agent (its usual role: S goes from +4 to +6, e.g. with $KMnO_4$, $K_2Cr_2O_7$, Cl_2) and an oxidizing agent towards stronger reducing agents such as H_2S (S goes from +4 to 0). The oxidation number +4 is intermediate, so it can move either way.

Bleaching action. SO_2 bleaches moist vegetable

2 SULPHURIC ACID: MANUFACTURE, PROPERTIES AND USES

In the previous section we explored the rich redox chemistry of sulphur, noting how it comfort-

ably adopts oxidation states from **−2** (in H_2S) through **+4** (in SO_2) to **+6** (in SO_3 and H_2SO_4). The **+6** state represents the fully oxidised, thermodynamically stable form of sulphur in the presence of oxygen. It is from this pinnacle that the single most important industrial chemical — sulphuric acid — is manufactured. A nation's annual consumption of H_2SO_4 is still regarded by economists as one of the most reliable barometers of its industrial development: it is the lifeblood of fertilizer production, petroleum refining, metal processing, and the manufacture of countless everyday materials. In this section we trace the journey from elemental sulphur to the concentrated acid, justify every condition in the Contact process with Le Chatelier's principle and kinetic theory, and then examine the remarkable trio of chemical personalities — acid, oxidiser, dehydrating agent — that make concentrated H_2SO_4 unique in the laboratory and in industry.

2.1 The Contact Process: Industrial Manufacture of Sulphuric Acid

The modern Contact process is a masterpiece of chemical engineering that converts sulphur (or sulphide ores) into sulphuric acid in three main stages, each optimised for yield, rate, and economic viability.

EXPERIMENT **Outline of the Contact Process**

The overall transformation can be summarised as:

1. Production of SO_2: Burning elemental sulphur or roasting sulphide ores (e.g. ZnS, FeS_2, Cu_2S) in excess dry air.
2. Catalytic oxidation to SO_3: $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$ $\quad \Delta H = -197\ \text{kJ mol}^{-1}$ (exothermic).
3. Absorption and dilution: SO_3 is absorbed in concentrated H_2SO_4 to form **oleum** ($H_2S_2O_7$), which is then diluted with water to give the desired acid concentration.

2.2 Step 1: Production and Purification of Sulphur Dioxide

PROPERTY **Sources of SO_2**

Elemental sulphur (recovered from oil and gas desulphurisation, or mined by the Frasch process) is burnt in a furnace with a controlled supply of dry air:

$$S(s) + O_2(g) \longrightarrow SO_2(g) \qquad \Delta H < 0$$

Alternatively, metal sulphide ores are roasted:

$$2\,ZnS(s) + 3\,O_2(g) \longrightarrow 2\,ZnO(s) + 2\,SO_2(g)$$

$$4\,FeS_2(s) + 11\,O_2(g) \longrightarrow 2\,Fe_2O_3(s) + 8\,SO_2(g)$$

The heat released is often recovered to generate steam for electricity or process heating.

WARNING **Purification is Critical — The Catalyst Poison**

The gas leaving the burner/roaster contains dust, arsenic oxides (As_2O_3), selenium, fluorine compounds, and moisture. Before entering the converter, it must be rigorously purified:

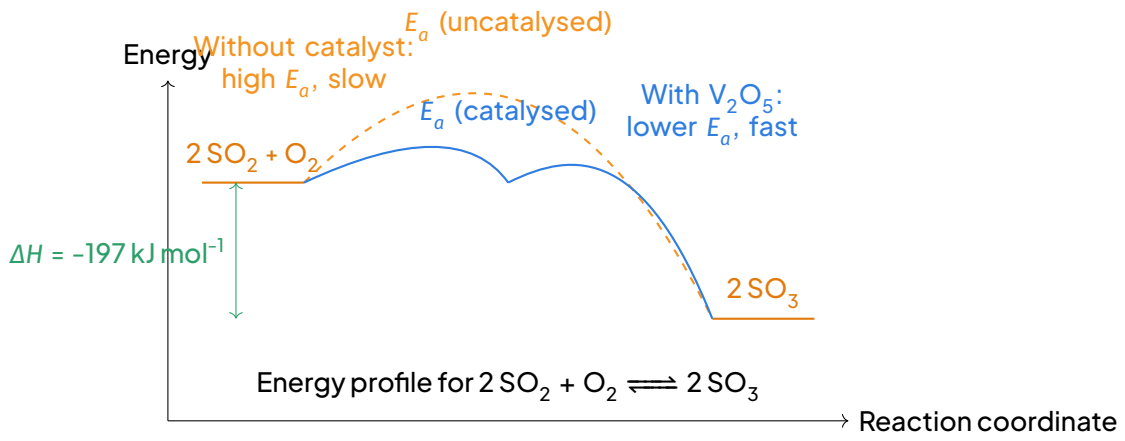
- Electrostatic precipitators / cyclones remove dust.
- Washing towers with water or acid spray remove soluble impurities and cool the gas.
- Drying towers with concentrated H_2SO_4 remove moisture (water vapour would form acid mist later and corrode equipment).
- Arsenic removal: Arsenic impurities poison the V_2O_5 catalyst irreversibly. The gas is passed through a bed of Fe_2O_3 or $Fe(OH)_3$ which absorbs As_2O_3 as ferric arsenate.

GCE Trap: A common exam question asks "Why is the gas dried?" Answer: To prevent corrosion and, crucially, to avoid formation of a stable H_2SO_4 mist in the absorber that would escape with the tail gases.

2.3 Step 2: Catalytic Oxidation of SO_2 to SO_3 — The Heart of the Process

The reversible, exothermic reaction is:

$$2SO_2(g) + O_2(g) \rightleftharpoons [V_2O_5][450^\circ C,\ 1-2\ atm]2SO_3(g) \qquad \Delta H = -197\ kJ\ mol^{-1}$$



Energy profile showing how the V_2O_5 catalyst provides an alternative reaction pathway with lower activation energy (E_a), speeding up both forward and reverse reactions equally without shifting the equilibrium position.

METHOD **Choosing Industrial Conditions — The Le Chatelier / Kinetics / Economics Balance**

The Contact process is the textbook example of optimising an industrial equilibrium. For $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$ ($\Delta H < 0, \Delta n_{gas} = -1$):

1. Temperature: Low temperature favours the forward reaction (exothermic → higher equilibrium yield). BUT too low a temperature makes the rate unacceptably slow even with a catalyst. Compromise: 400 °C to 450 °C. At 450 °C with V_2O_5, the equilibrium conversion is ≈ 97% and the rate is economically viable.
2. Pressure: High pressure favours the forward reaction (fewer gas moles on product side). BUT the equilibrium yield at 1 atm is already > 95% at 450 °C. Compressing large gas volumes is expensive (energy, thicker vessels). Decision: Operate at 1 atm to 2 atm (slightly above atmospheric to push gas through the plant). High pressure is not used.
3. Catalyst: Vanadium(V) oxide (V_2O_5) on a silica support. It lowers E_a dramatically, allowing the 450 °C compromise. It is cheap, resistant to poisoning (if gas is purified), and has a long life. Note: The catalyst does not change the equilibrium constant or the yield at equilibrium; it only helps the system reach equilibrium faster.
4. Concentration (Excess Oxygen): Using excess air (typically 2× stoichiometric O_2) shifts equilibrium to the right (Le Chatelier) and ensures near-complete conversion of the valuable SO_2. The unreacted gases are recycled.
5. Heat Management: The reaction is exothermic. The converter contains multiple catalyst beds with heat exchangers between them. The hot exit gas preheats the incoming feed, maintaining the optimal temperature and recovering energy.

DEFINITION **Oleum — Disulphuric(VI) Acid**

Oleum (fuming sulphuric acid) is a solution of sulphur trioxide in concentrated sulphuric acid, with the formula $H_2S_2O_7$. It is formed by the reaction:

$$SO_3(g) + H_2SO_4(l) \longrightarrow H_2S_2O_7(l)$$

Oleum is a stable, transportable intermediate. When water is added, it forms sulphuric acid:

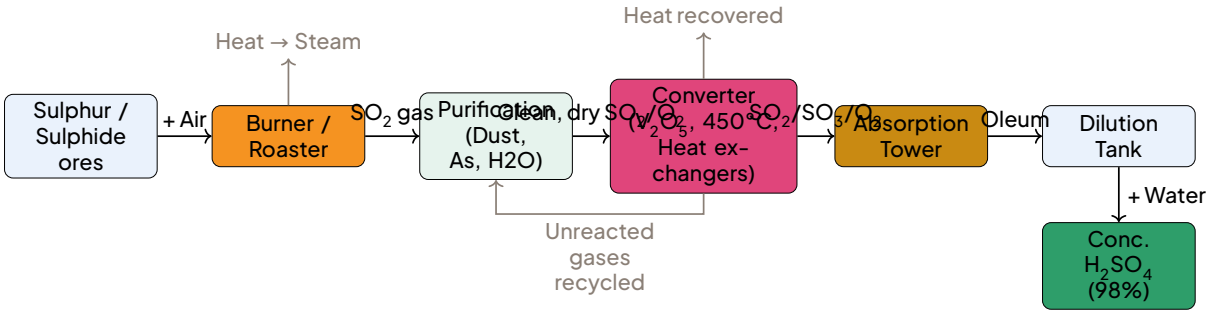
$$H_2S_2O_7(l) + H_2O(l) \longrightarrow 2\,H_2SO_4(l)$$

The percentage labelling of oleum (e.g. 20% oleum) refers to the mass of free SO_3 present per 100 g of oleum, or equivalently the mass of water needed to convert 100 g of oleum to 100% H_2SO_4.

2.4 Step 3: Absorption of SO_3 — Why Not Water Directly?

WARNING **The Mist Trap — Never Absorb SO_3 Directly in Water**

If SO_3 is bubbled into water, the reaction $SO_3 + H_2O \longrightarrow H_2SO_4$ is violently exothermic. The heat vaporises water and acid, creating a fine, stable acid mist (aerosol) that does not condense easily and escapes the absorber, causing pollution and product loss. Solution: SO_3 is absorbed in 98–99% concentrated H_2SO_4 in a packed absorption tower. The vapour pressure of water over concentrated H_2SO_4 is extremely low, so no mist forms. The product is oleum ($H_2S_2O_7$), which is then carefully diluted with water to the desired concentration (usually 98% for storage/transport).



Flow diagram of the Contact process. Key features: purification unit protects the catalyst; converter uses multiple V_2O_5 beds with inter-stage heat exchangers; absorption in concentrated acid avoids mist formation; final dilution yields 98% H_2SO_4.

2.5 Physical Properties of Concentrated Sulphuric Acid

KEY POINTS **Key Physical Properties of Conc. H_2SO_4 (98%)**

- Appearance: Colourless, dense, oily liquid (viscosity ≈ 27 mPa · s at 20 °C).
- Density: 1.84 g cm^{-3} — nearly twice as dense as water.
- Boiling point: 337 °C (decomposes slightly to $SO_3 + H_2O$ vapour).
- Hygroscopicity: Extremely strong affinity for water — excellent drying agent for gases (except NH_3 and H_2S, see below).
- Dilution: Highly exothermic ($\Delta H_{dil} \approx -80$ kJ mol^{-1} for infinite dilution). The heat released can boil water and spatter concentrated acid.

WARNING **Safe Dilution Technique — ALWAYS Acid into Water**

Rule: "Do as you oughta, add acid to water."

1. Use a heat-resistant container (Pyrex beaker).
2. Add the water first, then slowly pour the concentrated acid down the side of the beaker while stirring constantly.
3. The dense acid sinks, and the heat is dissipated into the larger volume of water.

Never pour water into concentrated acid — the water floats, boils violently at the interface, and ejects droplets of concentrated acid.

2.6 Chemical Properties: The Three Personalities of H_2SO_4

Sulphuric acid exhibits three distinct chemical behaviours depending on concentration and temperature. Recognising which role it is playing is essential for predicting products.

PROPERTY **Personality 1: Dilute H_2SO_4 — A Typical Dibasic Acid**

When dilute, H_2SO_4 behaves as a strong diprotic acid, donating two protons stepwise:

$$H_2SO_4(aq) \longrightarrow H^+(aq) + HSO_4^-(aq) \quad \text{(strong, complete)}$$

$$HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq) \quad (K_{a2} = 1.2 \times 10^{-2})$$

It undergoes all typical acid reactions:

- With reactive metals (above H in electrochemical series): $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2\uparrow$
- With bases (neutralisation): $2\,NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2\,H_2O$ (normal salt)
  $NaOH + H_2SO_4 \longrightarrow NaHSO_4 + H_2O$ (acid salt / hydrogensulphate)
- With carbonates/hydrogencarbonates: $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + CO_2\uparrow + H_2O$
- With metal oxides: $CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

Note: With $BaCl_2$ or $Pb(NO_3)_2$, dilute H_2SO_4 gives a white precipitate of $BaSO_4$ or $PbSO_4$ (insoluble sulphates) — a standard test for SO_4^{2-}.

PROPERTY **Personality 2: Hot Concentrated H_2SO_4 — An Oxidizing Agent**

When hot and concentrated, H_2SO_4 is reduced, with the sulphur oxidation state dropping from +6 to +4 (in SO_2) or lower. The acid provides both the oxidising power and the acidic medium.

Reducing Agent	Observation	Redox Half-Equations
Cu (s) (hot) Overall: $Cu + 2H_2SO_4(conc) \rightarrow [heat]CuSO_4 + SO_2 \uparrow + 2H_2O$	Blue solution (Cu^{2+}), colourless SO_2 gas	$Cu \longrightarrow Cu^{2+} + 2e^-$ $SO_4^{2-} + 4H^+ + 2e^- \longrightarrow SO_2 + 2H_2O$
C (s) (charcoal) Overall: $C + 2H_2SO_4(conc) \rightarrow [heat]CO_2 \uparrow + 2SO_2 \uparrow + 2H_2O$	SO_2 gas, CO_2 gas	$C + 2H_2O \longrightarrow CO_2 + 4H^+ + 4e^-$ $2SO_4^{2-} + 8H^+ + 4e^- \longrightarrow 2SO_2 + 4H_2O$
S (s) Overall: $S + 2H_2SO_4(conc) \rightarrow [heat]3SO_2 \uparrow + 2H_2O$	SO_2 gas (from both S and acid)	$S + 2H_2O \longrightarrow SO_2 + 4H^+ + 4e^-$

Key point: Cold concentrated H_2SO_4 does not oxidise copper. Heat is required to overcome the kinetic barrier.

PROPERTY **Personality 3: Concentrated H_2SO_4 — A Dehydrating Agent**

Concentrated H_2SO_4 has an immense affinity for water, stripping the elements of water (H and O in 2:1 ratio) from compounds. This is not an acid-base reaction; it is a dehydration.

- Charring of carbohydrates (sugar, glucose, starch, paper, wood):

$$C_{12}H_{22}O_{11}(s) \xrightarrow{\text{conc} \cdot H_2SO_4} 12\,C(s) + 11\,H_2O(l/g)$$

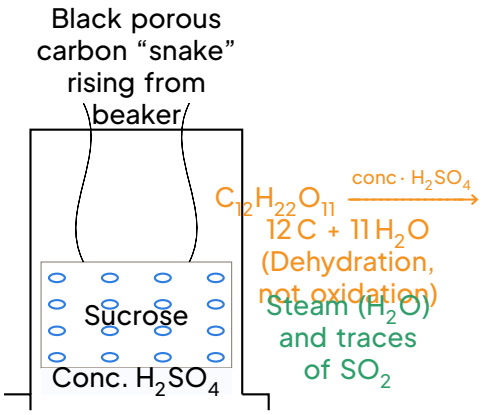
The black porous carbon "snake" rises from the beaker as steam and SO_2 (from slight oxidation) escape.

- Drying gases: Used in the laboratory to dry Cl_2, O_2, N_2, CO_2, SO_2, HCl, etc.

WARNING **Gases That CANNOT Be Dried with Conc. H_2SO_4**

1. Ammonia (NH_3): Acid-base reaction occurs: $NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$ or NH_4HSO_4. The gas is consumed.
2. Hydrogen sulphide (H_2S): Redox reaction occurs: $H_2S + H_2SO_4 \rightarrow S\downarrow + SO_2\uparrow + 2H_2O$. The gas is oxidised.
3. Alkenes (e.g. C_2H_4): Addition reaction / polymerisation occurs.

Alternative drying agents: CaO (for NH_3), P_4O_{10} or silica gel (for H_2S and general use).



The "black snake" demonstration: concentrated H_2SO_4 dehydrates sucrose, stripping H and O as water and leaving a rising column of elemental carbon. The reaction is highly exothermic, producing steam and a small amount of SO_2 from partial oxidation.

2.7 Uses of Sulphuric Acid

KEY POINTS **Major Industrial Uses of H_2SO_4**

1. Fertilizer manufacture (largest use, ~ 60%):
   - Superphosphate: $Ca_3(PO_4)_2 + 2\,H_2SO_4 \longrightarrow Ca(H_2PO_4)_2 + 2\,CaSO_4$ (single superphosphate, SSP).
   - Triple superphosphate: $Ca_3(PO_4)_2 + 3\,H_3PO_4 \longrightarrow 3\,Ca(H_2PO_4)_2$ (uses phosphoric acid made from H_2SO_4).
   - Ammonium sulphate: $2\,NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$.
2. Lead-acid storage batteries (car batteries): Electrolyte is ≈ 30% H_2SO_4 (4.2 mol dm^{-3}). Reaction: $PbO_2 + Pb + 2\,H_2SO_4 \rightleftharpoons 2\,PbSO_4 + 2\,H_2O$.
3. Paints and pigments: Manufacture of titanium dioxide (TiO_2, white pigment) via the sulphate process; production of barium sulphate (blanc fixe).
4. Detergents: Sulphonation of alkylbenzenes to make linear alkylbenzene sulphonates (LAS).
5. Metal processing: Pickling (removing oxide scale) of steel before galvanising or plating; leaching of copper and uranium ores.
6. Other: Rayon (viscose) production, explosives (nitration mixture with HNO_3), petroleum refining (alkylation catalyst), laboratory reagent.

2.8 GCE Exam Focus: Percentage Purity and Yield Calculations

Contact process calculations are a staple of Paper 2. They typically involve:

- Impure sulphur / sulphide ore → SO_2 (account for purity).
- Equilibrium conversion in converter (often given as % yield or calculated from K_p).
- Absorption efficiency.
- Final acid concentration / mass.

WORKED EXAMPLE 1 **Contact Process Yield Calculation**

Statement: 100 kg of iron pyrites (FeS_2) containing 85 % pure FeS_2 by mass is roasted in excess air. The SO_2 produced is converted to SO_3 in a Contact process converter with an equilibrium yield of 96 % (based on SO_2 converted). The SO_3 is absorbed in concentrated H_2SO_4 to form oleum, which is diluted to give 98 % H_2SO_4 by mass. Calculate the mass of 98 % acid produced. (Ar: Fe=56, S=32, O=16, H=1)

SOLUTION

Detailed Solution:

Step 1: Mass of pure FeS_2 $M_r(FeS_2) = 56 + 64 = 120\ g\ mol^{-1}$ Mass of pure $FeS_2 = 100\ kg \times 0.85 = 85\ kg = 85000\ g$

Step 2: Moles of FeS_2 and SO_2 produced Roasting: $4\,FeS_2 + 11\,O_2 \longrightarrow 2\,Fe_2O_3 + 8\,SO_2$ Mole ratio $FeS_2 : SO_2 = 4 : 8 = 1 : 2$ $n(FeS_2) = \frac{85000}{120} = 708.33\ mol$ $n(SO_2)$ (theoretical) $= 2 \times 708.33 = 1416.67\ mol$

Step 3: Moles of SO_3 formed (96% conversion) $2\,SO_2 + O_2 \rightleftharpoons 2\,SO_3$ Mole ratio $SO_2 : SO_3 = 1 : 1$ $n(SO_3) = 0.96 \times 1416.67 = 1360.0\ mol$

Step 4: Moles of H_2SO_4 formed Absorption + dilution: $SO_3 + H_2O \longrightarrow H_2SO_4$ (1:1 mole ratio) $n(H_2SO_4) = 1360.0\ mol$

Step 5: Mass of pure H_2SO_4 $M_r(H_2SO_4) = 2+32+64 = 98\ g\ mol^{-1}$ Mass pure $H_2SO_4 = 1360.0 \times 98 = 133280\ g = 133.28\ kg$

Step 6: Mass of 98 % acid Mass of 98% acid $= \frac{\text{mass pure acid}}{0.98} = \frac{133.28}{0.98} = 136.0\ kg$

Answer: 136 kg of 98 % H_2SO_4 (to 3 s.f.).

EXERCISE 1 **Practice: Sulphur Purity and Acid Yield**

Statement: A sample of crude sulphur of mass 50.0 kg and purity 92 % is burnt in air. The SO_2 is passed through a Contact process converter where the equilibrium conversion to SO_3 is 97.5 %. The SO_3 is absorbed in 98 % H_2SO_4 to form oleum, which is then diluted with water to produce 95 % H_2SO_4. Calculate the mass of 95 % acid obtained. Assume no losses in absorption or dilution.

SOLUTION **Exercise 1**

Solution:

1. Pure S mass = $50.0 \times 0.92 = 46.0\ kg = 46000\ g$.
2. $n(S) = 46000/32 = 1437.5\ mol$.
3. $S + O_2 \longrightarrow SO_2$: $n(SO_2) = 1437.5\ mol$.
4. $2\,SO_2 + O_2 \rightleftharpoons 2\,SO_3$ (1:1 ratio): $n(SO_3) = 0.975 \times 1437.5 = 1399.56\ mol$.
5. $SO_3 + H_2O \longrightarrow H_2SO_4$: $n(H_2SO_4) = 1399.56\ mol$.
6. Mass pure $H_2SO_4 = 1399.56 \times 98 = 137157\ g = 137.16\ kg$.
7. Mass 95 % acid $= 137.16/0.95 = 144.4\ kg$.

Answer: 144 kg (3 s.f.).

2.9 Summary: Key Points to Retain

KEY POINTS **Chapter Essentials — Sulphuric Acid**

- Contact Process: $S \rightarrow SO_2 \rightarrow [V_2O_5, 450^\circ C, 1-2atm]SO_3 \rightarrow [conc.H_2SO_4]H_2S_2O_7 \rightarrow [H_2O]H_2SO_4$.
- Conditions justified: 450 °C (yield/rate compromise), 1–2 atm (yield already high, compression costly), V_2O_5 catalyst (lowers E_a, no equilibrium shift), excess O_2 (shifts equilibrium right).
- Purification: Essential to remove catalyst poisons (As) and prevent mist (drying).
- Absorption: SO_3 into conc. H_2SO_4 (oleum) → dilution. Never direct into water (mist).
- Three roles: (1) Dilute = dibasic acid; (2) Hot conc. = oxidising agent (SO_2 evolution); (3) Conc. = dehydrating agent (charring, gas drying).
- Gases not dried by conc. H_2SO_4: NH_3 (acid-base), H_2S (redox), alkenes (addition).
- Safe dilution: Acid into water, slowly, with stirring.
- Uses: Fertilizers (superphosphate, $(NH_4)_2SO_4$), batteries, pigments, detergents, metal pickling.
- Calculations: Track moles through each stage; apply % purity at start, % yield at converter, % concentration at end.

DID YOU KNOW? **Sulphuric Acid in Cameroon**

Cameroon does not currently produce sulphuric acid on a large industrial scale; it is imported for use in the petroleum refinery (SONARA) for alkylation, in the aluminum smelter (ALUCAM) for anode production, and crucially for the manufacture of phosphate fertilizers (e.g. at the proposed Kribi fertilizer plant). The National Fertilizer Subsidy Programme distributes NPK fertilizers whose phosphate component ultimately derives from sulphuric acid attack on phosphate rock. Thus, while the Contact process towers may not dot the Cameroonian landscape, the acid's fingerprints are on every bag of fertilizer that boosts maize yields in the North-West or cocoa in the South-West.

3 OXIDES OF SULPHUR AND ATMOSPHERIC POLLUTION

In the previous section we saw that the Contact process deliberately converts SO_2 into SO_3 and then into H_2SO_4 inside a carefully controlled plant. Exactly the same chemistry occurs, uncontrolled, in the open air whenever sulphur-containing fuels are burned. The atmosphere becomes an accidental chemical reactor, and the product — sulphuric acid — falls back to Earth

in the rain. This section studies that unwanted twin of the Contact process: the formation of sulphur oxides, their transformation into **acid rain**, their effects on health and the environment, and the technologies used to limit the damage.

3.1 Sources of Sulphur Dioxide

DEFINITION **Sulphur dioxide**

Sulphur dioxide, SO_2, is a colourless, choking gas in which sulphur has the oxidation state +4. It is produced whenever elemental sulphur or sulphur compounds are burned in air:

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

The main sources of SO_2 in the atmosphere are:

- **Burning of fossil fuels.** Coal and fuel oil contain sulphur (as organic sulphur compounds and as pyrite, FeS_2) at levels typically between 0.5% and 5% by mass. On combustion this sulphur is oxidised quantitatively to SO_2. Thermal power stations and industrial boilers are therefore major emitters.
- **Roasting of sulphide ores.** In the extraction of metals such as copper and zinc, sulphide ores are roasted in air, releasing SO_2:

  $$2\,ZnS(s) + 3\,O_2(g) \longrightarrow 2\,ZnO(s) + 2\,SO_2(g)$$

  In modern smelters this gas is captured and fed to a Contact plant — a profitable way of combining metal extraction with sulphuric acid manufacture.
- **Volcanic emissions.** Volcanoes inject very large quantities of SO_2 directly into the atmosphere. This is a natural source that no human policy can control; on a global scale it is comparable to industrial emissions.
- **Gas flaring and refining.** Crude oil and natural gas contain sulphur compounds (notably H_2S). In the Cameroonian context, the refinery at Limbe (SONARA), gas flaring associated with offshore oil production, and thermal power stations burning fuel oil all contribute sulphur oxides to the local atmosphere.

DID YOU KNOW? **Sulphur in Cameroon**

Natural gas from the Logbaba and Sanaga Sud fields contains hydrogen sulphide, H_2S, which must be removed before the gas is used — both because H_2S is extremely toxic and because burning sulphurous gas would release SO_2. The removed sulphur can itself be recovered and sold to sulphuric acid plants.

3.2 From SO_2 to SO_3: Atmospheric Oxidation

SO_2 does not remain unchanged in the air. Over hours to days it is oxidised to SO_3 by two main routes:

1. **Photo-oxidation (gas phase).** Solar ultraviolet radiation initiates a radical chain reaction. The hydroxyl radical $^{\bullet}OH$ (formed from ozone photolysis) attacks SO_2:

$$SO_2(g) + {}^{\bullet}OH(g) \longrightarrow HOSO_2^{\bullet}(g)$$

The $HOSO_2^{\bullet}$ radical reacts rapidly with O_2 to give SO_3 and $HO_2^{\bullet}$, which regenerates $^{\bullet}OH$. The overall stoichiometry is:

$$2SO_2(g) + O_2(g) \rightarrow [h\equiv]2SO_3(g)$$

2. **Catalytic oxidation on particulate matter.** Soot, fly ash and metal-rich dust particles in polluted air provide surfaces on which SO_2 and O_2 react much faster. Water droplets containing dissolved metal ions (e.g. Fe^{3+}, Mn^{2+}) also catalyse the oxidation of dissolved SO_2 to H_2SO_4. This heterogeneous route is often dominant in humid, smoky air.

Notice the parallel with the Contact process: there, a V_2O_5 catalyst at about 450 °C performs the same oxidation efficiently; in the atmosphere, dust and sunlight do it slowly and for free.

3.3 Acid Rain: Formation and pH

PROPERTY **Natural acidity of rainwater**

Even in the absence of any pollution, rainwater is slightly acidic, with pH ≈ 5.6, because it dissolves atmospheric carbon dioxide to form the weak acid carbonic acid:

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

Acidity caused by SO_2 and nitrogen oxides is additional to this natural background.

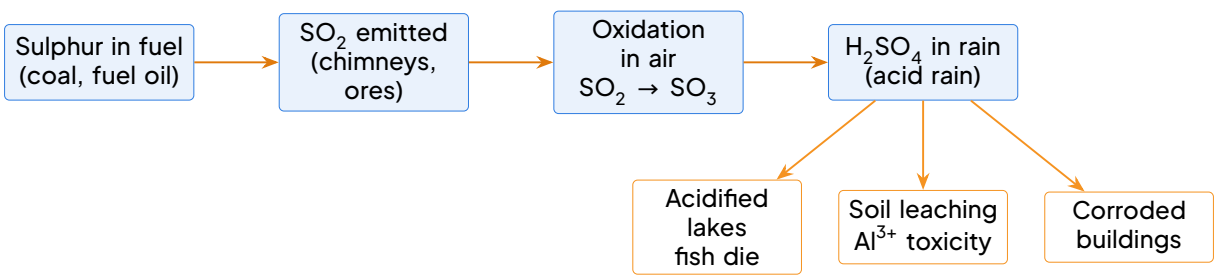
Both sulphur oxides are acid anhydrides: they react with water to give acids.

$SO_2(g) + H_2O(l) \longrightarrow H_2SO_3(aq)$	sulphurous acid (weak, diprotic)
$SO_3(g) + H_2O(l) \longrightarrow H_2SO_4(aq)$	sulphuric acid (strong, diprotic)

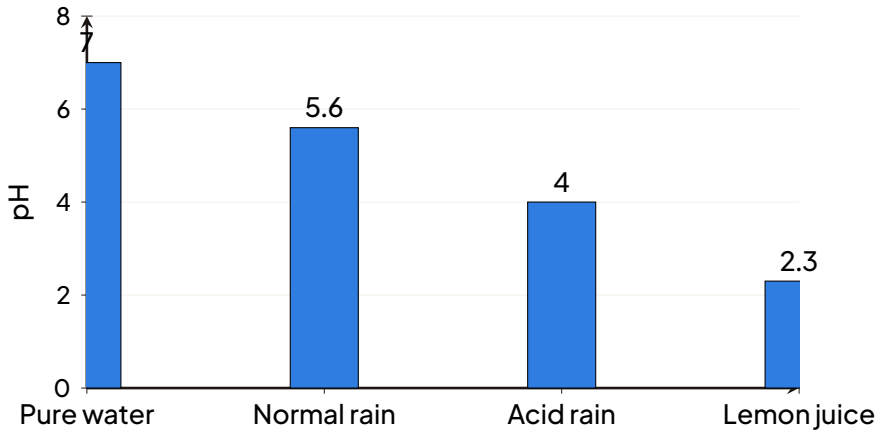
H_2SO_4 is a strong acid and dissociates extensively, so even small concentrations of SO_3 lower the pH of rain dramatically. Since pH is logarithmic, a fall from 5.6 to 4.0 means the concentration of H^+ has been multiplied by about 40.

DEFINITION **Acid rain**

Acid rain is precipitation (rain, snow, fog or dew) with a pH below about 5.0, caused by dissolved sulphur oxides (SO_2, SO_3) and nitrogen oxides (NO, NO_2) which form sulphurous, sulphuric, nitrous and nitric acids in atmospheric water.



From sulphur in fuel to environmental damage: the acid-rain pathway.



Comparison of pH: pure water (7), normal rain (≈ 5.6), acid rain (≈ 4) and lemon juice (≈ 2.3).

3.4 Environmental Effects of Acid Rain

Acidification of lakes and aquatic life.

When acid rain drains into lakes with little natural buffering (e.g. lakes on granitic bedrock), the pH of the water falls. Below pH 5, fish eggs fail to hatch and the gills of adult fish are damaged; below pH 4.5, most fish species disappear and the lake becomes biologically impoverished — clear, blue and almost lifeless.

Soil leaching and aluminium toxicity.

Acid rain dissolves and washes away (leaches) essential soil cations such as Ca^{2+}, Mg^{2+} and K^+, impoverishing the soil. Worse, the extra H^+ attacks clay minerals and releases Al^{3+} ions into soil water:

$$Al(OH)_3(s) + 3\,H^+(aq) \longrightarrow Al^{3+}(aq) + 3\,H_2O(l)$$

Al^{3+} is toxic to plant roots and to fish: it damages root membranes, blocks nutrient uptake, and in gills it causes mucus clogging. Forest decline observed in heavily polluted regions is largely attributed to this combination of nutrient loss and aluminium toxicity.

Corrosion of buildings and metals.

Limestone and marble (both $CaCO_3$) react directly with the acid:

$$CaCO_3(s) + H_2SO_4(aq) \longrightarrow CaSO_4(s) + H_2O(l) + CO_2(g)$$

The surface of statues and carved facades literally dissolves, and fine detail is lost. Iron and steel structures are attacked too, with the acid accelerating rusting — a real economic cost for bridges, roofs and railway lines.

Damage to crops and forests.

Acid deposition damages the waxy cuticle of leaves, leaches nutrients from foliage, and weakens trees so that they succumb more easily to drought, frost, insects and disease. Crop yields fall on acidified soils unless lime is regularly added.

3.5 Direct Health Effects of SO_2 and Secondary Pollutants

SO_2 itself is a powerful respiratory irritant. It dissolves in the moisture lining the airways to form sulphurous acid, causing coughing, constriction of the bronchi and, after prolonged exposure, chronic **bronchitis**. Asthmatics are particularly sensitive: even short exposures to elevated SO_2 levels can trigger severe attacks.

In addition, SO_2 is converted in the atmosphere into **sulphate aerosols** — tiny suspended particles of $(NH_4)_2SO_4$ and H_2SO_4 droplets. These secondary pollutants:

- scatter sunlight and reduce **visibility** (the haze over industrial regions);
- penetrate deep into the lungs when inhaled, contributing to respiratory and cardiovascular disease;
- can remain airborne for days and travel hundreds of kilometres, exporting pollution far from its source.

WARNING **Greenhouse effect ≠ acid rain**

Do not confuse the greenhouse effect (caused mainly by CO_2 and CH_4, leading to global warming) with acid rain (caused by SO_2 and nitrogen oxides). They are different problems with different gases, different chemistries and different remedies. Examiners penalise answers that mix the two.

3.6 Control Measures

1. Use of low-sulphur fuels. Switching from high-sulphur coal or fuel oil to natural gas or low-sulphur coal reduces emissions at the source.

2. Hydrodesulphurization of fuels. In refineries, sulphur compounds are removed from liquid fuels by reaction with hydrogen over a catalyst; the sulphur leaves as H_2S, which is then converted to elemental sulphur (Claus process) and sold.

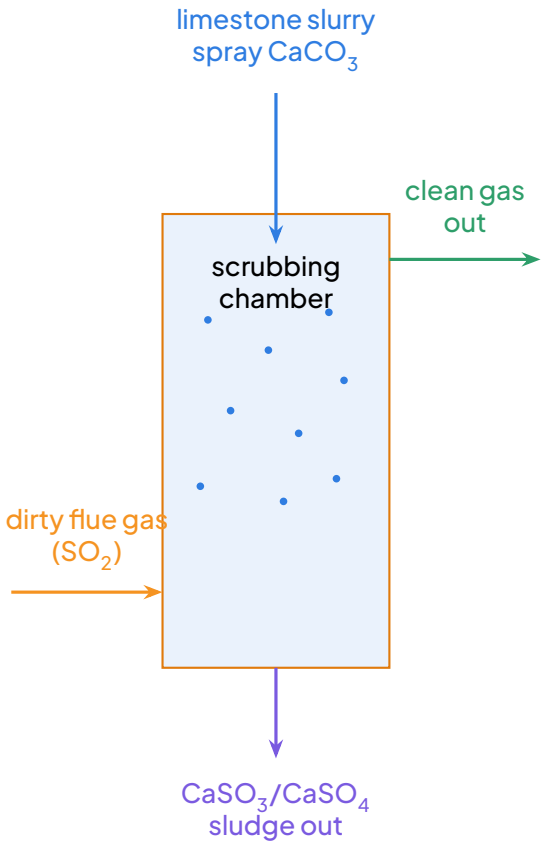
3. Flue-gas desulphurization (FGD). The most important end-of-pipe technology: the exhaust gases are "scrubbed" with a slurry of limestone ($CaCO_3$) or lime (CaO) before release.

METHOD **Flue-gas desulphurization: the chemistry**

The scrubber neutralises the acidic SO_2 with a basic calcium compound:

1. With limestone:
$$CaCO_3(s) + SO_2(g) \longrightarrow CaSO_3(s) + CO_2(g)$$
2. With quicklime:
$$CaO(s) + SO_2(g) \longrightarrow CaSO_3(s)$$
3. The calcium sulphite is then usually oxidised with air to gypsum, a saleable product used in plasterboard:
$$2\,CaSO_3(s) + O_2(g) \longrightarrow 2\,CaSO_4(s)$$

Up to 90% of the SO_2 can be removed this way.



Schematic of a limestone wet scrubber: flue gas enters, slurry spray absorbs SO_2, clean gas leaves.

4. Tall chimneys. Building very tall chimneys carries the plume high into the atmosphere so that

ground-level concentrations near the plant are reduced.

WARNING **Tall chimneys disperse, they do not remove**

A common mistake is to claim that tall chimneys "reduce pollution". They do not reduce the total SO_2 emitted: the same mass of gas is released, merely spread over a wider area and carried further downwind — sometimes across national borders. Dispersal is not removal; only scrubbing or cleaner fuels actually cut emissions.

WORKED EXAMPLE 2 **Sulphur dioxide from burning fuel oil**

A thermal power station burns 1000 kg of fuel oil per hour containing 2.0% sulphur by mass. Calculate (a) the mass of SO_2 produced per hour, assuming all the sulphur is converted; (b) the mass of limestone, $CaCO_3$, needed per hour to scrub this SO_2 completely. (A_r: C = 12, O = 16, S = 32, Ca = 40.)

SOLUTION

(a) Mass of SO_2.
Mass of sulphur burned per hour:

$$m(\mathrm{S}) = 1000 \times \frac{2.0}{100} = 20\ \mathrm{kg}$$

From $S + O_2 \longrightarrow SO_2$, one mole of S gives one mole of SO_2.

$$n(\mathrm{S}) = \frac{20\,000\ \mathrm{g}}{32\ \mathrm{g\ mol^{-1}}} = 625\ \mathrm{mol}$$

$$m(\mathrm{SO_2}) = 625 \times (32 + 2 \times 16) = 625 \times 64 = 40\,000\ \mathrm{g} = 40\ \mathrm{kg}$$

(b) Mass of limestone.
From $CaCO_3 + SO_2 \longrightarrow CaSO_3 + CO_2$, one mole of $CaCO_3$ removes one mole of SO_2.

$$n(\mathrm{CaCO_3}) = 625\ \mathrm{mol}$$

$$m(\mathrm{CaCO_3}) = 625 \times (40 + 12 + 3 \times 16) = 625 \times 100 = 62\,500\ \mathrm{g} = 62.5\ \mathrm{kg}$$

So the station emits 40 kg of SO_2 per hour and would need 62.5 kg of pure limestone per hour to scrub it all.

DID YOU KNOW? **A GCE extra: nitrogen oxides also make acid rain**

Sulphur is not the only culprit. Nitrogen oxides (NO and NO_2, together called NO_x), produced in vehicle engines and lightning, dissolve in rain to give nitric acid, HNO_3. In many

industrialised countries NO_x now contributes as much to acid rain as SO_2. We shall meet these oxides again when we study nitrogen compounds and fertilisers in the next sections.

EXERCISE 2 **Checking understanding**

1. Explain why "pure" unpolluted rainwater is not neutral.
2. Write balanced equations for the formation of sulphuric acid in the atmosphere starting from sulphur in coal.
3. State two environmental effects and one health effect of sulphur dioxide pollution.
4. A smelter roasts zinc blende according to $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$. Calculate the volume of SO_2 at s.t.p. released per tonne (10^6 g) of ZnS roasted. (Molar volume at s.t.p. = 22.4 $dm^3\,mol^{-1}$; A_r: Zn = 65, S = 32.)

SOLUTION **Exercise n°1**

1. Rainwater dissolves atmospheric CO_2, forming the weak acid H_2CO_3; hence its pH is about 5.6 rather than 7.
2. $S + O_2 \longrightarrow SO_2$; $2\,SO_2 + O_2 \longrightarrow 2\,SO_3$; $SO_3 + H_2O \longrightarrow H_2SO_4$.
3. Environmental: acidification of lakes killing fish; corrosion of limestone buildings (or soil leaching, forest damage). Health: respiratory irritation, bronchitis, aggravation of asthma.
4. $M(ZnS) = 65 + 32 = 97\ g\,mol^{-1}$; $n(ZnS) = 10^6/97 \approx 10\,309$ mol. From the equation, $n(SO_2) = n(ZnS)$, so $V(SO_2) = 10\,309 \times 22.4 \approx 2.31 \times 10^5\ dm^3$ at s.t.p.

KEY POINTS **Key points**

- SO_2 comes from burning sulphur-containing fossil fuels, roasting sulphide ores, volcanoes and gas flaring.
- In air, SO_2 is oxidised (photo-oxidation via $^{\bullet}OH$ radicals, catalysis on particles) to SO_3; the oxides dissolve in rain to give H_2SO_3 and H_2SO_4.
- Normal rain has pH ≈ 5.6 (dissolved CO_2); acid rain has pH < 5.
- Effects: acidified lakes, soil leaching with Al^{3+} toxicity, corrosion of $CaCO_3$ buildings

and metals, crop and forest damage, respiratory illness, sulphate aerosol haze.

- Control: low-sulphur fuels, hydrodesulphurization, flue-gas scrubbing ($CaCO_3$ + $SO_2 \longrightarrow CaSO_3 + CO_2$). Tall chimneys only disperse pollution.
- Never confuse acid rain (SO_2, NO_x) with the greenhouse effect (CO_2, CH_4).

4 NITROGEN, AMMONIA AND NITRIC ACID

In the previous section we saw how sulphur, burned in fuels, escapes into the atmosphere as SO_2 and returns to Earth as acid rain. But sulphur is not the only non-metal whose chemistry shapes our environment. Every breath you take is about 78 % nitrogen gas, N_2 — yet this enormous reservoir is almost useless to plants and animals in its elemental form. Turning inert N_2 into ammonia and nitric acid is one of the greatest achievements of industrial chemistry, and it feeds the world through fertilisers. In this section we study nitrogen itself, its two most important compounds — ammonia NH_3 and nitric acid HNO_3 — and the two industrial processes (Haber and Ostwald) that the GCE examiners love to test.

4.1 Nitrogen in Nature and its Oxidation States

An inert but abundant gas. Air contains about 78 % N_2 by volume. Nitrogen is remarkably unreactive at ordinary temperatures: it does not burn, does not support combustion, and reacts with very few reagents directly. The reason is structural. The two nitrogen atoms are joined by a **triple bond**, N ≡ N, with a very high bond dissociation energy (about 945 kJ mol^{-1}). Breaking this bond requires a large activation energy, so reactions of N_2 are slow even when they are energetically favourable. This is why nitrogen fixation — natural or industrial — always demands drastic conditions (lightning, high temperature and pressure, or specialised enzymes).

The nitrogen cycle in outline. Although N_2 is inert, nitrogen atoms circulate continuously between the atmosphere, the soil and living organisms:

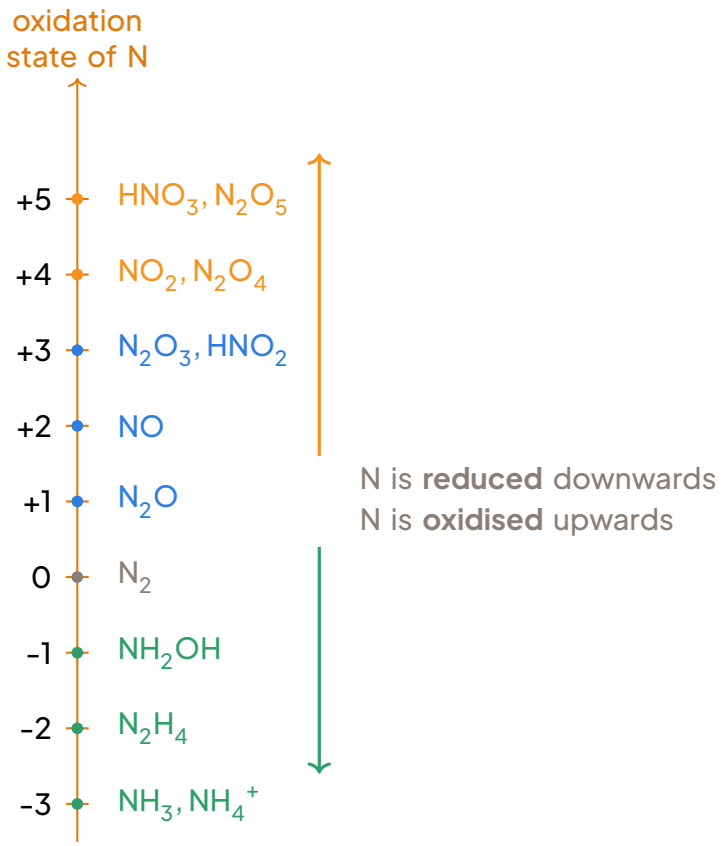
- **Nitrogen fixation:** conversion of N_2 into nitrogen compounds. Naturally by lightning ($N_2 + O_2 \longrightarrow 2\,NO$ at very high temperature, then $NO \longrightarrow NO_2 \longrightarrow HNO_3$ washed into the soil as nitrates) and by Rhizobium bacteria in the root nodules of legumes (beans, groundnuts — well known to Cameroonian farmers). Industrially by the Haber process.
- **Nitrification:** soil bacteria (Nitrosomonas, Nitrobacter) oxidise ammonium compounds first to nitrites NO_2^-, then to nitrates NO_3^-, which plants absorb.
- **Assimilation and decay:** plants convert nitrates into proteins; animals eat the plants; decomposition and excretion return ammonium compounds to the soil.
- **Denitrification:** other bacteria reduce nitrates back to N_2, completing the cycle.

DEFINITION **Nitrogen fixation**

Nitrogen fixation is the conversion of atmospheric nitrogen gas, N_2, into nitrogen compounds (such as NH_3, NO or nitrates) that plants can absorb and use. It occurs naturally (lightning discharges; Rhizobium bacteria in legume root nodules) and industrially (the Haber process).

Oxidation states from -3 **to** +5. Like sulphur, nitrogen displays a wide range of oxidation states — in fact an even wider one, every integer value from -3 to +5. You must be able to assign the oxidation state of N in any compound using the rules (O is -2, H is +1).

Oxidation state	Compound	Name
-3	NH_3, NH_4^+	ammonia, ammonium ion
-2	N_2H_4	hydrazine
-1	NH_2OH	hydroxylamine
0	N_2	nitrogen gas
+1	N_2O	dinitrogen monoxide
+2	NO	nitrogen monoxide
+3	N_2O_3, HNO_2	dinitrogen trioxide, nitrous acid
+4	NO_2, N_2O_4	nitrogen dioxide
+5	HNO_3, N_2O_5, NO_3^-	nitric acid, dinitrogen pentoxide, nitrate



Oxidation-state ladder of nitrogen, from −3 to +5, with the important oxides and oxyacids.

WARNING **Assigning oxidation states**

In N_2O, do not write N = +2 by treating it like NO. Solve $2x + (-2) = 0$, giving $x = +1$ (an average value). Similarly in NH_4^+: $x + 4(+1) = +1$, so $x = -3$. Always use the overall charge of the species.

4.2 Ammonia: Laboratory Preparation

Ammonia, NH_3, is a colourless gas with a characteristic pungent, choking smell (think of smelling salts or stale urine, where it forms by decomposition of urea). In the laboratory it is prepared by heating an **ammonium salt with a base** — classically ammonium chloride with calcium hydroxide (slaked lime):

$$2\,NH_4Cl(s) + Ca(OH)_2(s) \xrightarrow{\text{heat}} CaCl_2(s) + 2\,NH_3(g) + 2\,H_2O(l)$$

The mixture is heated gently in a hard-glass test tube tilted slightly downwards (so that the water produced does not run back onto the hot glass). Ammonia is **lighter than air** and **extremely soluble in water**, so it is collected by **upward delivery** (downward displacement of air) — never

over water.

METHOD **Drying ammonia**

Ammonia is a basic gas, so it must be dried with a basic drying agent: quicklime, CaO, placed in a drying tower.

1. Do not use concentrated H_2SO_4: it is acidic and reacts: $2\,NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$.
2. Do not use anhydrous $CaCl_2$: it combines with ammonia to form an addition compound, $CaCl_2 \cdot 8\,NH_3$.
3. Do not use P_2O_5 (also acidic).
4. Use CaO: it is basic and absorbs water without reacting with NH_3.

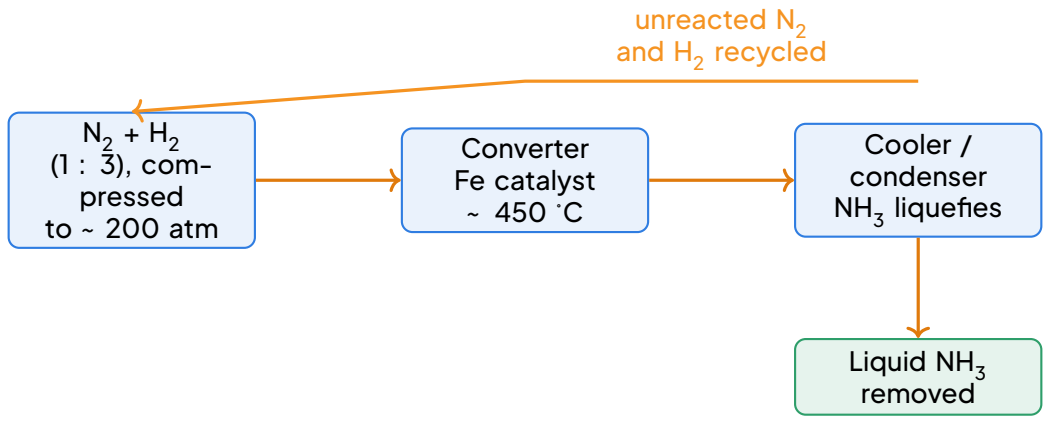
4.3 The Haber Process: Industrial Manufacture of Ammonia

The Haber process is the industrial answer to nitrogen fixation — and arguably the most important chemical process ever invented, since the fertilisers it enables feed billions of people.

Raw materials. Nitrogen is obtained by fractional distillation of liquid air. Hydrogen is obtained from natural gas by steam reforming: $CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3\,H_2(g)$.

The equilibrium. The synthesis is a reversible, exothermic reaction:

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \qquad \Delta H = -92\ \text{kJ mol}^{-1}$$



Flow diagram of the Haber process, with the recycle loop for unreacted gases.

Choosing the conditions — a compromise. Apply Le Chatelier's principle to $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$:

- **Pressure:** 4 moles of gas → 2 moles of gas, so high pressure shifts equilibrium towards NH_3

and also increases the rate. About 200 atm is used; higher pressures give better yields but the cost of compression and of building safe plant rises sharply.

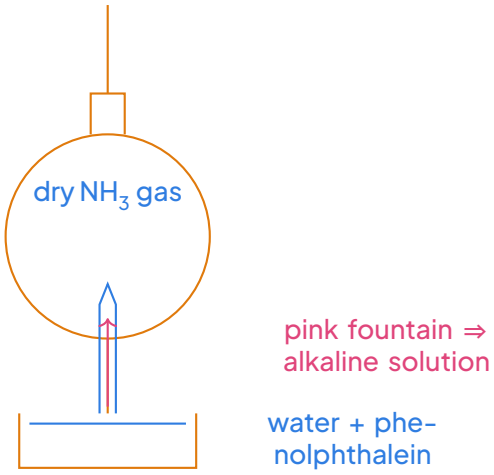
- **Temperature:** the forward reaction is exothermic, so low temperature favours the yield of NH_3. But at low temperature the reaction is extremely slow (the strong N ≡ N bond gives a huge activation energy). A temperature of about 450 °C is chosen as a **compromise** between yield and rate.
- **Catalyst:** finely divided **iron** speeds up attainment of equilibrium without changing the position of equilibrium.
- **Removal of product:** the gas mixture leaving the converter contains only about 15 % NH_3; it is cooled so that ammonia liquefies and is removed, and the unreacted N_2/H_2 are **recycled**, giving an overall conversion of about 98 %.

WARNING **The classic Haber trap**

Lowering the temperature would give a higher equilibrium yield of NH_3 — but the reaction would be far too slow to be economic. 450 °C is not the temperature of maximum yield; it is a compromise between equilibrium yield and reaction rate. Saying "the catalyst increases the yield" is also wrong: a catalyst increases only the rate, never the equilibrium yield.

4.4 Properties of Ammonia

Extreme solubility and alkalinity — the fountain experiment. Ammonia is one of the most soluble gases known: about 700 volumes of NH_3 dissolve in one volume of water at room temperature. This is dramatically shown by the **fountain experiment**. A round-bottomed flask is filled with dry ammonia and fitted with a jet tube dipping into water containing a little phenolphthalein. When a few drops of water enter the flask, the ammonia dissolves, the pressure inside collapses, and water is driven up through the jet as a spectacular fountain — turning **pink**, proving that the solution is alkaline.



The ammonia fountain experiment: dissolution of NH_3 lowers the pressure and water jets in, turning pink with phenolphthalein.

The solution is alkaline because ammonia is a **weak base**, partially accepting a proton from water:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

PROPERTY **Ammonia as a Lewis base**

The nitrogen atom in NH_3 carries a lone pair of electrons, which it can donate to a proton or to a metal ion. Ammonia is therefore a **Lewis base** (and a Brønsted base). This single fact explains both its alkalinity in water and its ability to form complex ions with transition-metal cations. (No formal proof is examinable; you must be able to use the idea.)

Precipitation of metal hydroxides. Because aqueous ammonia supplies OH^- ions, it precipitates insoluble metal hydroxides from salt solutions — a key tool in qualitative analysis:

$$Cu^{2+}(aq) + 2\,OH^-(aq) \longrightarrow Cu(OH)_2(s) \quad \text{pale blue precipitate}$$
$$Fe^{3+}(aq) + 3\,OH^-(aq) \longrightarrow Fe(OH)_3(s) \quad \text{red-brown precipitate}$$

Complex formation. With excess ammonia solution, some hydroxide precipitates redissolve as ammine complexes. The classic case is copper(II):

$$Cu(OH)_2(s) + 4\,NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2\,OH^-(aq)$$

The deep blue colour of tetraamminecopper(II), $[Cu(NH_3)_4]^{2+}$, is a confirmatory test for Cu^{2+}. (With silver chloride, ammonia forms $[Ag(NH_3)_2]^+$, dissolving the precipitate — another useful test.)

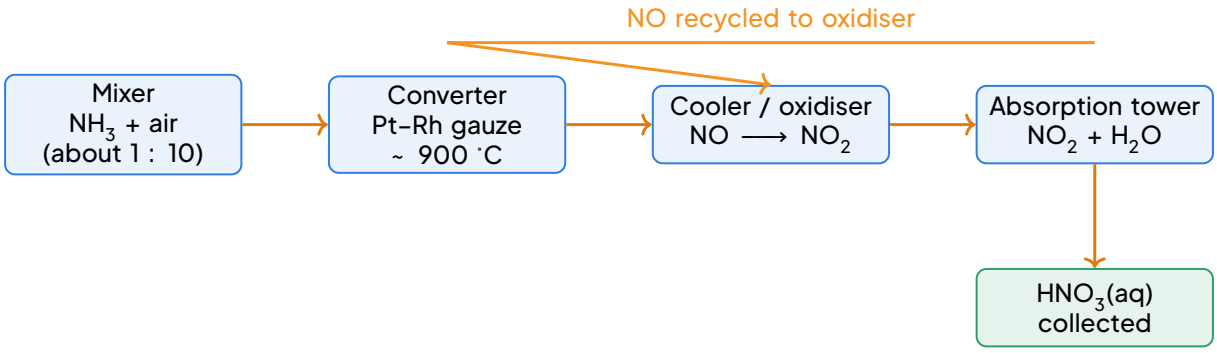
Ammonia as a reducing agent. Nitrogen in NH_3 is in its lowest oxidation state (-3), so it can only be oxidised: ammonia is a reducing agent.

- It reduces heated copper(II) oxide: $2\,NH_3(g) + 3\,CuO(s) \longrightarrow 3\,Cu(s) + N_2(g) + 3\,H_2O(l)$ (black CuO turns brown-pink).
- It burns in oxygen (and is oxidised catalytically over platinum): $4\,NH_3(g) + 3\,O_2(g) \longrightarrow 2\,N_2(g) + 6\,H_2O(g)$; over hot Pt the product is mainly NO — the basis of the Ostwald process below.

Uses of ammonia. The great majority (about 80 %) goes into **fertiliser manufacture** (ammonium salts, urea, nitrates). It is also used as a **refrigerant** gas in large cold stores and in household **cleaning agents**, and as the raw material for nitric acid.

4.5 The Ostwald Process: Manufacture of Nitric Acid

Nitric acid is made by the catalytic oxidation of ammonia — elegantly linking the two compounds of this section.



Flow diagram of the Ostwald process for the manufacture of nitric acid.

1. **Catalytic oxidation of ammonia.** A mixture of ammonia and excess air passes over a platinum–rhodium gauze at about 900 °C:

$$4NH_3(g) + 5O_2(g) \rightarrow [Pt/Rh,\ 900\,^{\circ}C]\,4NO(g) + 6H_2O(g) \qquad \Delta H < 0$$

   The gauze glows red-hot: the reaction is exothermic and, once started, maintains its own temperature.

2. **Oxidation of NO.** The gases are cooled and the nitrogen monoxide reacts with more oxygen:

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

3. **Absorption in water.** The nitrogen dioxide is absorbed in water in the presence of excess air:

$$3\,NO_2(g) + H_2O(l) \longrightarrow 2\,HNO_3(aq) + NO(g)$$

   The NO produced is **recycled** and re-oxidised, so the overall yield is very high. The acid obtained (about 60 %) is concentrated by distillation.

4.6 Properties of Nitric Acid

A strong acid. Pure nitric acid is a colourless fuming liquid. In water it is fully ionised:

$$HNO_3(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + NO_3^-(aq)$$

It shows all the typical reactions of a strong acid:

$HNO_3(aq) + NaOH(aq) \longrightarrow NaNO_3(aq) + H_2O(l)$	base
$2\,HNO_3(aq) + CuO(s) \longrightarrow Cu(NO_3)_2(aq) + H_2O(l)$	basic oxide
$2\,HNO_3(aq) + Na_2CO_3(aq) \longrightarrow 2\,NaNO_3(aq) + H_2O(l) + CO_2(g)$	carbonate

An oxidising acid. Here lies the crucial difference from hydrochloric or dilute sulphuric acid. The nitrate group, with nitrogen in its maximum oxidation state +5, is itself a powerful oxidising agent. Consequently, when nitric acid attacks metals, it is the **nitrate that is reduced**, not the H^+ — so **hydrogen is generally not evolved**.

With copper, the products depend on concentration:

concentrated: $Cu(s) + 4\,HNO_3(conc) \longrightarrow Cu(NO_3)_2(aq) + 2\,NO_2(g) + 2\,H_2O(l)$
(brown fumes of NO_2; solution turns blue-green)

dilute: $3\,Cu(s) + 8\,HNO_3(dil) \longrightarrow 3\,Cu(NO_3)_2(aq) + 2\,NO(g) + 4\,H_2O(l)$
(colourless NO, which browns in air as it forms NO_2)

WARNING **No hydrogen with nitric acid!**

A very common GCE mistake is to write $Cu + 2\,HNO_3 \longrightarrow Cu(NO_3)_2 + H_2$. This is wrong. Nitric acid is an oxidising acid: with dilute acid copper gives NO, with concentrated acid it gives NO_2 — never hydrogen (except with very dilute acid and very reactive metals such as Mg or Mn, a rare exception). Learn the two copper equations above; they are set again and again.

Passivation. Concentrated nitric acid renders aluminium and iron **passive**: a thin, impermeable layer of oxide forms on the metal surface and protects it from further attack. This is why concentrated HNO_3 can be transported in aluminium tanks.

Decomposition in light. Concentrated nitric acid slowly decomposes, especially when warm or exposed to light:

$$4\,HNO_3(aq) \xrightarrow{\text{light}} 4\,NO_2(g) + O_2(g) + 2\,H_2O(l)$$

The dissolved brown NO_2 gives old nitric acid its characteristic **yellow colour**. For this reason the acid is stored in brown bottles.

4.7 Worked Examples and GCE Examination Focus

WORKED EXAMPLE 3 **Redox half-equations for the reduction of nitric acid**

Write the balanced half-equation, in acid solution, for the reduction of the nitrate ion (a) to NO_2, (b) to NO. State the change in oxidation state of nitrogen in each case.

SOLUTION

Balance O with H_2O, then H with H^+, then charge with electrons.
(a) $NO_3^- \longrightarrow NO_2$. Add one H_2O on the right for the extra O, then $2H^+$ on the left:

$$NO_3^-(aq) + 2\,H^+(aq) + e^- \longrightarrow NO_2(g) + H_2O(l)$$

Nitrogen goes from +5 to +4: a gain of one electron.
(b) $NO_3^- \longrightarrow NO$. Add $2H_2O$ on the right, then $4H^+$ on the left:

$$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \longrightarrow NO(g) + 2\,H_2O(l)$$

Nitrogen goes from +5 to +2: a gain of three electrons.
Combining (b) with the oxidation of copper, $Cu \longrightarrow Cu^{2+} + 2\,e^-$ (multiplied by 3, while (b) is multiplied by 2), regenerates the full equation $3\,Cu + 8\,HNO_3 \longrightarrow 3\,Cu(NO_3)_2 + 2\,NO + 4\,H_2O$ — an excellent check.

WORKED EXAMPLE 4 **Yield calculation on the Haber process**

A converter is fed with 100 mol of N_2 and 300 mol of H_2. At equilibrium the mixture contains 20 % conversion of the nitrogen. Calculate (a) the amount of ammonia formed, (b) the total number of moles of gas at equilibrium.

SOLUTION

(a) N_2 reacted = 0.20×100 = 20 mol. From $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$, ammonia formed = 2×20 = $\boxed{40\text{ mol}}$.
(b) H_2 reacted = 3×20 = 60 mol. Remaining: N_2: 100−20 = 80 mol; H_2: 300−60 = 240 mol; NH_3: 40 mol.

$$n_{\text{total}} = 80 + 240 + 40 = \boxed{360\text{ mol}}$$

Note that the total has decreased from 400 to 360 mol — consistent with 4 volumes of reactants giving only 2 volumes of product.

WORKED EXAMPLE 5 **The Ostwald process from ammonia to nitric acid**

Calculate the maximum mass of nitric acid that can be obtained from 1.70 kg of ammonia, assuming quantitative conversion. ($M(NH_3) = 17$, $M(HNO_3) = 63$.)

SOLUTION

Combining the three steps, each nitrogen atom of NH_3 ends up in one molecule of HNO_3 (the NO by-product is recycled), so the overall mole ratio is $NH_3 : HNO_3 = 1 : 1$.

$$n(NH_3) = \frac{1700}{17} = 100\ \text{mol} \quad \Rightarrow \quad n(HNO_3) = 100\ \text{mol}$$

$$m(HNO_3) = 100 \times 63 = 6300\ \text{g} = \boxed{6.30\ \text{kg}}$$

In practice, yields of about 96 % are achieved thanks to the recycling of NO.

EXERCISE 3 **Test yourself**

1. Assign the oxidation state of nitrogen in N_2O, NO, N_2O_3, NO_2 and HNO_3.
2. Explain, with equations, why ammonia cannot be dried with concentrated sulphuric acid or anhydrous calcium chloride.
3. State the conditions used in the Haber process and justify each one using Le Chatelier's principle and kinetic arguments.
4. Write balanced equations for the three steps of the Ostwald process.
5. A blue solution of copper(II) sulphate is treated with a few drops of ammonia solution, then with excess. Describe and explain all observations, with equations.

SOLUTION **Exercise n° 1**

1. N_2O: +1; NO: +2; N_2O_3: +3; NO_2: +4; HNO_3: +5.
2. H_2SO_4 is acidic: $2\,NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$. $CaCl_2$ forms an addition compound: $CaCl_2 + 8\,NH_3 \longrightarrow CaCl_2 \cdot 8\,NH_3$. Only a basic dryer such as CaO is suitable.
3. Fe catalyst (increases rate, not yield); ~ 450 °C (compromise: lower T favours yield of the exothermic reaction but is too slow); ~ 200 atm (fewer moles of gas on the product side, so high pressure increases both rate and yield); NH_3 liquefied off and unreacted gases recycled.
4. $4\,NH_3 + 5\,O_2 \xrightarrow{Pt/Rh} 4\,NO + 6\,H_2O$; $2\,NO + O_2 \longrightarrow 2\,NO_2$; $3\,NO_2 + H_2O \longrightarrow 2\,HNO_3 + NO$.

5. A few drops: pale blue precipitate of $Cu(OH)_2$: $Cu^{2+} + 2\,OH^- \longrightarrow Cu(OH)_2$. Excess ammonia: precipitate dissolves to a deep blue solution of the complex $[Cu(NH_3)_4]^{2+}$: $Cu(OH)_2 + 4\,NH_3 \longrightarrow [Cu(NH_3)_4]^{2+} + 2\,OH^-$.

KEY POINTS **Key points of this section**

- N_2 is inert because of the strong N ≡ N triple bond; nitrogen shows every oxidation state from −3 (NH_3) to +5 (HNO_3).
- Laboratory NH_3: heat NH_4Cl with $Ca(OH)_2$; dry over CaO; collect by upward delivery.
- Haber process: $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$, $\Delta H = -92\,kJ\,mol^{-1}$; Fe catalyst, ~ 450 °C, ~ 200 atm — a compromise between rate and yield.
- NH_3 is a weak base and a Lewis base: it precipitates metal hydroxides, forms complexes such as deep blue $[Cu(NH_3)_4]^{2+}$, and acts as a reducing agent.
- Ostwald process: $NH_3 \longrightarrow NO \longrightarrow NO_2 \longrightarrow HNO_3$ over Pt–Rh at ~ 900 °C, with NO recycled.
- HNO_3 is a strong acid and an oxidising agent: with Cu it gives NO_2 (conc.) or NO (dil.), never H_2; it passivates Al and Fe and turns yellow in light due to dissolved NO_2.

DID YOU KNOW? **Fritz Haber: feeding the world**

Fritz Haber developed the ammonia synthesis in Germany in the early 20th century and received the Nobel Prize in Chemistry for it. It has been estimated that a large fraction of the nitrogen atoms in the human body today passed through a Haber plant — the fertilisers that grew our food were made from ammonia. Every bag of NPK or urea fertiliser used on the maize farms of the North-West and West Regions of Cameroon carries a little of this chemistry. In the next section we look closely at these fertilisers — and at what happens when too much of them washes into our rivers and lakes.

5 FERTILIZERS AND THEIR ENVIRONMENTAL EFFECTS

In the previous section we followed nitrogen from the inert gas in the atmosphere through the Haber process to ammonia, and on through the Ostwald process to nitric acid. We now ask the obvious question: why does industry make millions of tonnes of ammonia and nitric acid every year? The answer, in one word, is **fertilizers**. The same chemistry that feeds the world, however, can also poison rivers, lakes and drinking water if it is misused. This final section closes the loop:

we see why crops need fertilizers, how the main fertilizers are made from the very chemicals of this chapter, how to compare them quantitatively, and what happens when they escape into the environment.

5.1 Why Fertilizers Are Needed

Walk through any intensively farmed zone of Cameroon — the banana plantations of the Littoral, the maize fields of the West Region — and you will see the same story. Natural soil contains the mineral nutrients plants need, but every harvest removes nutrients in the crop and carries them away. Unlike a natural ecosystem, where dead leaves and organisms decompose and return their minerals to the soil, a farm exports its nutrients to the market. After a few seasons of continuous cropping, yields fall dramatically unless the lost nutrients are replaced.

Three elements are needed in the largest amounts and are therefore the most commonly deficient. They are remembered by the letters **N**, **P** and **K**.

Nutrient	Role in the plant	Deficiency symptom
Nitrogen (N)	Component of proteins, chlorophyll and nucleic acids; drives leaf and stem growth	Yellowing of older leaves (chlorosis), stunted growth
Phosphorus (P)	Part of ATP (energy transfer), DNA and cell membranes; promotes root development and flowering	Poor root growth, purplish leaves, delayed maturity
Potassium (K)	Regulates water balance and stomata; activates enzymes; improves disease and drought resistance	Scorched leaf edges, weak stems, poor fruit quality

DEFINITION **Fertilizer**

A **fertilizer** is any substance, natural or manufactured, added to soil or water to supply one or more of the nutrient elements essential for plant growth, principally nitrogen, phosphorus and potassium.

5.2 Types of Fertilizers

Straight fertilizers supply a single major nutrient. The important examples for this syllabus are:

- **Ammonium nitrate**, NH_4NO_3 — a nitrogen fertilizer, highly soluble, fast acting.
- **Ammonium sulphate**, $(NH_4)_2SO_4$ — supplies nitrogen and sulphur; useful on sulphur-deficient soils.
- **Urea**, $CO(NH_2)_2$ — the most concentrated solid nitrogen fertilizer in common use.
- **Superphosphate** — a mixture of $Ca(H_2PO_4)_2$ and $CaSO_4$, the main phosphorus fertilizer; the calcium dihydrogenphosphate is water-soluble and therefore available to plants, whereas the original phosphate rock is not.

Compound (NPK) fertilizers contain two or three of the major nutrients blended in one product, e.g. a "15–15–15" or "20–10–10" grade.

WARNING **What the NPK numbers really mean**

The three numbers on a fertilizer bag give the percentages by mass of N, P_2O_5 and K_2O equivalents — not the percentages of elemental phosphorus and potassium. A 20–10–10 fertilizer contains 20 % N, 10 % P_2O_5 equivalent and 10 % K_2O equivalent. This is a classic GCE trap: never convert the label directly into elemental P or K without being asked to.

5.3 Manufacture: Linking the Industrial Chemistry of this Chapter

Notice how the fertilizers are made directly from the products of the Haber, Ostwald and Contact processes — this is why those three processes dominate the chemical industry.

Ammonium nitrate is made by neutralising ammonia with nitric acid (the Ostwald product):

$$NH_3(g) + HNO_3(aq) \longrightarrow NH_4NO_3(aq)$$

followed by evaporation of the water and prilling (forming small pellets from droplets of the melt) to give the solid fertilizer.

Ammonium sulphate is made by neutralising ammonia with sulphuric acid (the Contact product):

$$2\,NH_3(g) + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

Superphosphate is made by treating insoluble phosphate rock, $Ca_3(PO_4)_2$, with sulphuric acid:

$$Ca_3(PO_4)_2(s) + 2\,H_2SO_4(aq) \longrightarrow Ca(H_2PO_4)_2(s) + 2\,CaSO_4(s)$$

The product mixture (soluble $Ca(H_2PO_4)_2$ + gypsum, $CaSO_4$) is sold directly as "single superphosphate". The chemistry is simply an acid–base displacement: the stronger acid (H_2SO_4) protonates the phosphate to give the soluble dihydrogenphosphate.

5.4 Comparing Fertilizers: Percentage Nitrogen

A farmer (and an examiner!) wants to know which fertilizer delivers the most nitrogen per franc. The measure is the **percentage by mass of nitrogen** in the compound.

METHOD **Calculating the percentage of nitrogen in a fertilizer**

1. Write the formula and count the number of nitrogen atoms per formula unit.
2. Calculate the mass of nitrogen in one mole: n(N atoms) × 14.
3. Calculate the molar mass M of the compound.
4. Apply: $\%N = \dfrac{\text{mass of N per mole}}{M} \times 100$.

WORKED EXAMPLE 6 **Percentage nitrogen in ammonium nitrate and urea**

Calculate the percentage by mass of nitrogen in (a) ammonium nitrate, NH_4NO_3; (b) urea, $CO(NH_2)_2$. Hence state which supplies more nitrogen per tonne. (A_r: H = 1, C = 12, N = 14, O = 16.)

SOLUTION

(a) Ammonium nitrate, NH_4NO_3. The formula contains 2 nitrogen atoms (one in NH_4^+, one in NO_3^-).

$$M(NH_4NO_3) = 14 + 4(1) + 14 + 3(16) = 80\ \text{g mol}^{-1}$$

$$\%N = \frac{2 \times 14}{80} \times 100 = \frac{28}{80} \times 100 = \boxed{35\,\%}$$

(b) Urea, $CO(NH_2)_2$. The formula contains 2 nitrogen atoms.

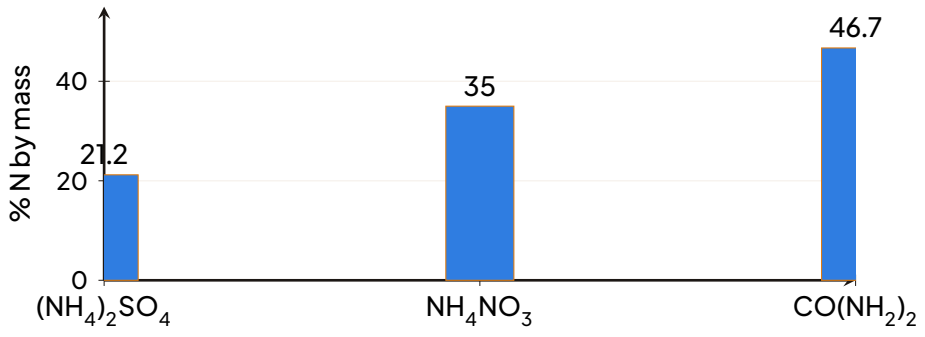
$$M(CO(NH_2)_2) = 12 + 16 + 2\,[14 + 2(1)] = 60\ \text{g mol}^{-1}$$

$$\%N = \frac{2 \times 14}{60} \times 100 = \frac{28}{60} \times 100 \approx \boxed{46.7\,\%}$$

Conclusion: urea is the more concentrated source of nitrogen — one tonne of urea supplies about 467 kg of N against 350 kg for one tonne of ammonium nitrate. (For comparison, $(NH_4)_2SO_4$ gives $\frac{28}{132} \times 100 \approx 21.2\,\%$.)

WARNING **Count the nitrogen atoms first!**

The most common error is forgetting that NH_4NO_3 contains two nitrogen atoms and writing $\%N = \frac{14}{80} \times 100 = 17.5\,\%$. Always count atoms before computing.



Percentage of nitrogen by mass in the three common straight nitrogen fertilizers.

5.5 Eutrophication: When Fertilizers Reach the Water

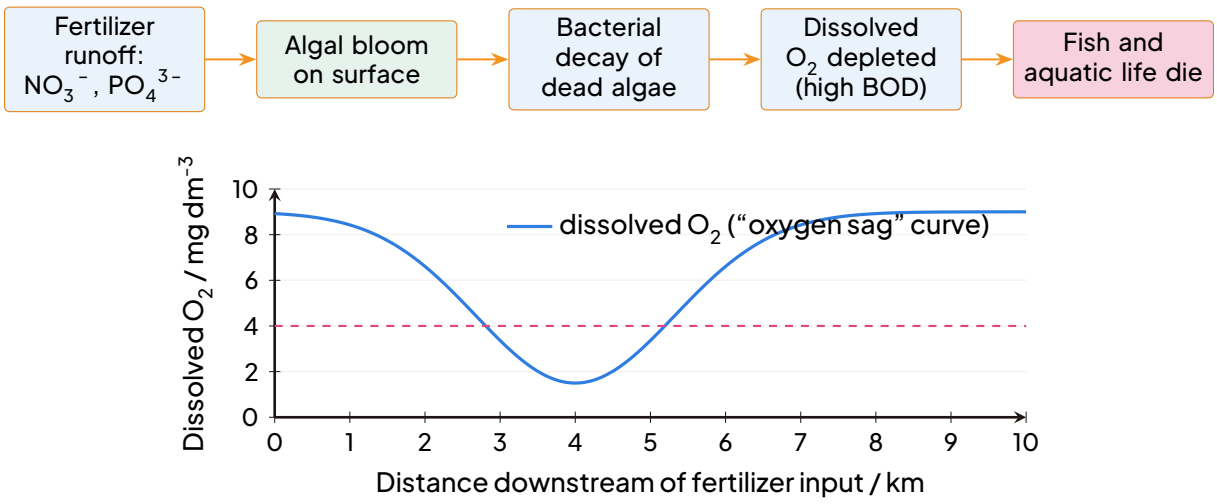
Fertilizer spread on a field does not all stay there. Nitrates, NO_3^-, are extremely soluble and are not held by soil particles, so rain leaches them through the soil into streams; phosphates are carried mainly in surface runoff with eroded soil. The nutrients end up in rivers and lakes — exactly the wrong place.

DEFINITION **Eutrophication**

Eutrophication is the excessive enrichment of a body of water with plant nutrients, especially nitrates and phosphates, leading to explosive algal growth, depletion of dissolved oxygen and the death of aquatic life.

The chain of events is a beautiful — and tragic — piece of applied biology and chemistry:

1. **Nutrient input:** nitrates and phosphates from fertilizer runoff (also from sewage and detergents) enter the water.
2. **Algal bloom:** algae, suddenly well fed, multiply explosively and cover the surface as a green carpet.
3. **Light blockage and death:** the dense surface layer blocks sunlight; algae and plants below die.
4. **Aerobic decomposition:** bacteria decompose the dead organic matter, consuming dissolved oxygen as they respire. The water develops a high **biochemical oxygen demand** (BOD).
5. **Oxygen depletion:** dissolved O_2 falls so low that fish and other aquatic animals suffocate and die. Anaerobic decay then releases foul gases such as H_2S and CH_4.



The eutrophication sequence (top) and the characteristic "oxygen sag": dissolved O_2 plunges just downstream of the nutrient input, then slowly recovers as the organic load is used up.

DID YOU KNOW? **Eutrophication in Cameroon**

The Municipal Lake in Yaoundé has repeatedly suffered algal blooms and fish deaths linked to nutrient-rich runoff from surrounding farms, gardens and untreated waste water. Similar problems affect streams draining intensively fertilised plantations and market-gardening zones. Eutrophication is not a distant, textbook problem — it is visible in our own cities.

5.6 Other Environmental Effects of Fertilizers

Nitrate in drinking water. Leached nitrate can contaminate wells and boreholes. In infants, gut bacteria reduce NO_3^- to nitrite, NO_2^-, which oxidises the iron(II) of haemoglobin to iron(III), forming methaemoglobin, which cannot carry oxygen. The result is **methaemoglobinaemia**, the "blue baby syndrome". This is why drinking-water standards set a strict limit on nitrate concentration.

Soil acidification. Long-term use of ammonium-based fertilizers slowly acidifies the soil.

PROPERTY **Acidifying action of ammonium fertilizers**

Nitrifying bacteria in the soil oxidise ammonium ions to nitrate, releasing hydrogen ions:

$$NH_4^+(aq) + 2\,O_2(g) \xrightarrow{\text{nitrifying bacteria}} NO_3^-(aq) + 2\,H^+(aq) + H_2O(l)$$

The released H^+ lowers the soil pH over the years, so farmers must periodically add lime (CaO or $CaCO_3$) to neutralise the acidity. (The overall process is examinable as an explanation, not a proof.)

Gaseous emissions from soils. Fertilised soils emit nitrogen oxides. Of particular concern is **dinitrogen monoxide**, N_2O (nitrogen oxidation state +1), released by microbial nitrification and denitrification. N_2O is a powerful **greenhouse gas** — far more effective per molecule than CO_2 — and it also contributes to the destruction of stratospheric ozone.

5.7 Oxides of Nitrogen and Photochemical Smog

Fertilizers are not the only nitrogen pollution. This completes the picture begun with the oxides of sulphur in Section 3.

Sources of NO and NO_2 (together called NO_x):

- **Vehicle engines:** at the very high temperatures of combustion, atmospheric nitrogen and oxygen combine:

$$N_2(g) + O_2(g) \rightarrow [\text{high T}]2NO(g) \qquad 2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

- **Lightning:** the enormous energy of a lightning flash drives the same $N_2 + O_2$ reaction — nature's own nitrogen fixation.

Photochemical smog. In strong sunlight over traffic-choked cities, NO_2 absorbs light and splits:

$$NO_2(g) \xrightarrow{\text{sunlight}} NO(g) + O(g) \qquad O(g) + O_2(g) \longrightarrow O_3(g)$$

The ozone and atomic oxygen react with unburnt hydrocarbons to give a brown, irritating haze — **photochemical smog** — containing harmful **tropospheric ozone** and compounds that attack eyes and lungs. Note the irony: ozone high in the stratosphere protects us, but ozone at ground level is a pollutant.

Control: catalytic converters. Modern exhaust systems pass the gases over a platinum/rhodium catalyst, which reduces the nitrogen oxides back to harmless nitrogen while oxidising carbon monoxide and unburnt hydrocarbons:

$$2\,NO(g) + 2\,CO(g) \xrightarrow{Pt/Rh} N_2(g) + 2\,CO_2(g)$$

> **WARNING** **Eutrophication is not acid rain**
>
> Do not confuse these two pollution problems. Acid rain is caused by SO_2 and NO_x gases dissolving in rain to form acids; it is controlled by flue-gas desulphurisation and catalytic converters. Eutrophication is caused by dissolved nutrients (NO_3^-, PO_4^{3-}) in water; it is controlled by careful fertilizer use and sewage treatment. Different causes, different victims, different remedies — examiners reward candidates who keep them apart.

5.8 Sustainable Use of Fertilizers

The goal is not to abandon fertilizers — without them, Cameroon could not feed its growing population — but to use them wisely:

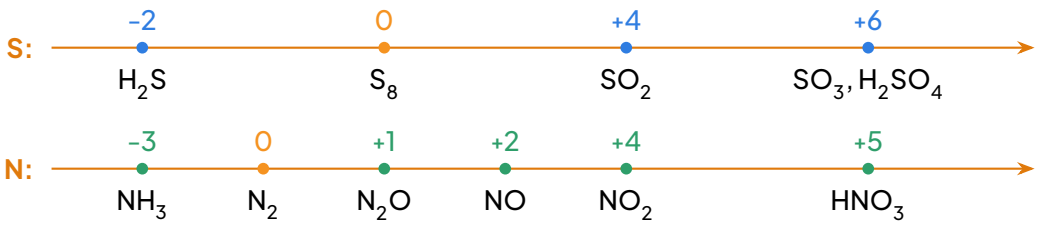
- **Controlled application rates:** apply only what the crop can absorb, guided by soil testing; excess fertilizer is wasted money and guaranteed pollution.

- **Correct timing:** avoid spreading just before heavy rain, which washes nutrients straight into streams.
- **Organic manure and compost:** release nutrients slowly, improve soil structure, and recycle farm waste.
- **Legume crop rotation:** beans, groundnuts and soya host Rhizobium bacteria in root nodules that fix atmospheric N_2 naturally, reducing the need for nitrogen fertilizer in the following season.
- **Buffer strips:** bands of unfertilised grass or trees along rivers and lakes trap runoff nutrients and sediment before they reach the water.

5.9 Chapter Synthesis: Sulphur and Nitrogen in Perspective

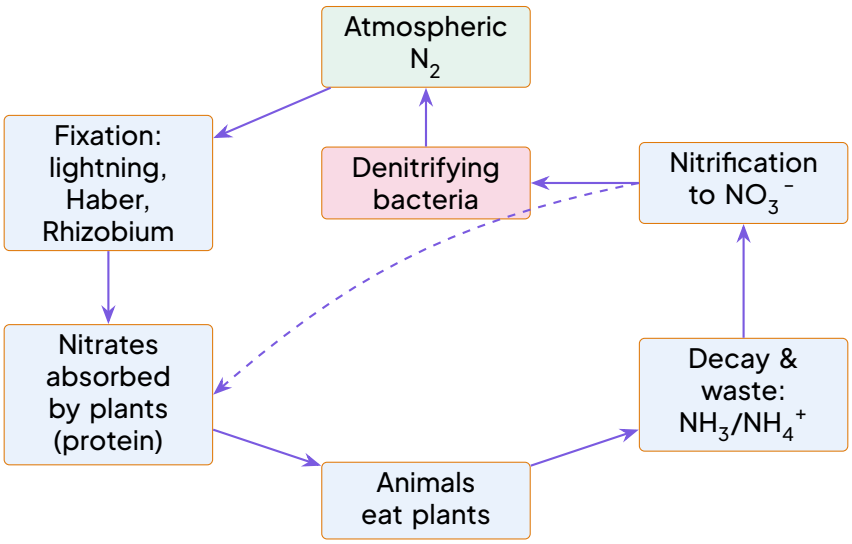
We close the chapter by gathering its two great themes.

Oxidation-state maps. Both elements display a wide range of oxidation states, and the chemistry of this chapter is largely the chemistry of moving between them:



Oxidation-state maps for the key sulphur and nitrogen species of this chapter. Oxidation (loss of electrons) moves to the right; reduction moves to the left.

The nitrogen cycle ties fixation, life, decay and return to the atmosphere into one grand loop:



The nitrogen cycle: fixation converts N_2 into usable forms; nitrification regenerates nitrate; denitrification returns N_2 to the atmosphere.

The three industrial processes that anchor the chapter:

Process	Product	Key equation	Conditions
Haber	NH_3	$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$	~450 °C, ~200 atm, Fe catalyst
Ostwald	HNO_3	$4\,NH_3(g) + 5\,O_2(g) \longrightarrow 4\,NO(g) + 6\,H_2O(g)$, then $NO \longrightarrow NO_2 \longrightarrow HNO_3$	~900 °C, Pt–Rh catalyst
Contact	H_2SO_4	$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$; $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$; $H_2S_2O_7 + H_2O \longrightarrow 2\,H_2SO_4$	~450 °C, V_2O_5 catalyst

KEY POINTS **Key points of this section**

- Crops remove N, P and K from the soil; fertilizers replace them. N → leaves/protein, P → roots/energy, K → water balance/disease resistance.
- $\%N = \dfrac{\text{mass of N in formula}}{M} \times 100$: NH_4NO_3 = 35 %, urea ≈ 46.7 %, $(NH_4)_2SO_4$ ≈ 21.2 %.
- NPK labels give %N, $\%P_2O_5$ and $\%K_2O$ equivalents.
- Eutrophication: runoff nutrients → algal bloom → aerobic decay → O_2 depletion (high BOD) → fish death.

- Nitrate in drinking water causes blue-baby syndrome; ammonium fertilizers acidify soil ($NH_4^+ \longrightarrow NO_3^- + 2\,H^+$); soils emit the greenhouse gas N_2O.
- NO_x from engines and lightning cause photochemical smog and tropospheric ozone; catalytic converters reduce NO back to N_2.
- Sustainable practice: right rate, right timing, manure, legume rotation, buffer strips.

EXERCISE 4 **Fertilizer value and the environment**

(a) Calculate the percentage by mass of nitrogen in ammonium sulphate, $(NH_4)_2SO_4$. (A_r: H = 1, N = 14, O = 16, S = 32.)
(b) A farmer applies ammonium nitrate at 200 kg per hectare. What mass of nitrogen is applied per hectare?
(c) Explain, with the aid of one equation, why years of ammonium sulphate use make the soil acidic, and name a substance used to correct this.
(d) Describe the sequence of events by which fertilizer runoff can kill fish in a lake, and state two farming practices that reduce this risk.

SOLUTION **Exercise 1**

(a) $M((NH_4)_2SO_4) = 2(14) + 8(1) + 32 + 4(16) = 132\ \text{g mol}^{-1}$; mass of N = $2 \times 14 = 28$.

$$\%N = \frac{28}{132} \times 100 \approx 21.2\,\%$$

(b) NH_4NO_3 is 35 % N, so mass of N = $0.35 \times 200 = 70$ kg per hectare.
(c) Nitrifying bacteria oxidise ammonium to nitrate, releasing hydrogen ions:

$$NH_4^+(aq) + 2\,O_2(g) \longrightarrow NO_3^-(aq) + 2\,H^+(aq) + H_2O(l)$$

The accumulating H^+ lowers the pH. It is corrected by adding lime (CaO or $CaCO_3$), a base that neutralises the acid.
(d) Nitrates and phosphates leach or run off into the lake; the nutrient enrichment causes an algal bloom; when the algae die, aerobic bacteria decompose them and consume dissolved oxygen (high BOD); the oxygen concentration falls below the level fish need, and they suffocate. Risk is reduced by: applying only the amount of fertilizer the crop needs; timing application to avoid heavy rain; keeping buffer strips of vegetation along waterways; using organic manure or legume rotation. (Any two.)

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

This integration activity brings together the whole chapter in a single, realistic decision-making problem. By the end of the activity you should be able to:

- **Calculate** the percentage by mass of nitrogen in common fertilizers and compare their **cost per kilogram of nitrogen**;
- **Write balanced equations** for the manufacture of urea, ammonium sulphate and an NPK fertilizer, tracing each back to the **Haber process**, the **Ostwald process** and the **Contact process**, with their operating conditions;
- **Explain chemically** how fertilizer runoff causes **eutrophication** and fish deaths, and why ammonium sulphate can **acidify** a soil;
- **Construct and justify** a costed fertilization plan together with realistic pollution-control measures, and defend your recommendation orally.

The activity is designed for group work (3–4 students), followed by a 5-minute presentation and class debate. It reproduces the kind of evidence-based reasoning expected in the GCE Advanced Level structured and essay questions.

METHOD **How to compare fertilizer costs**

1. Calculate the molar mass of the fertilizer.
2. Find the mass of nitrogen per mole (or per 100 g).
3. Compute %N = (mass of N / molar mass) × 100.
4. Mass of N per bag = bag mass × %N/100.
5. Cost per kg N = bag price / mass of N per bag.
6. Always compare per kg of nutrient, never per bag.

6.2 Situation

Case Study: Fertilizer Choice and Pollution Control for a Ndop Plain Rice Cooperative.

The Ndop Plain in the North West Region of Cameroon is one of the country's major rice-producing areas. The Mboghaa Rice Farmers' Cooperative cultivates 50 ha of paddy rice. Yields have fallen over recent seasons, so the cooperative commissioned a soil analysis and collected market prices for three fertilizers available in Bamenda. At the same time, fishermen on a stream draining the plain reported sudden **fish deaths** in August, shortly after heavy rains that followed the main fertilizer application.

Data pack 1 – Soil analysis (per hectare, top 20 cm).

Parameter	Measured value	Agronomist's comment
Available nitrogen	Low	Deficient for rice; supplement needed
Available phosphorus (P)	Low	Deficient; supplement needed
Available potassium (K)	Adequate	No supplement needed
Soil pH	5.6	Slightly acidic; avoid further acidification

The agronomist recommends applying 80 kg of nitrogen and 40 kg of P_2O_5 equivalent per hectare for the season.

Data pack 2 – Fertilizers on the Bamenda market (50 kg bags).

Fertilizer	Composition	Price per 50 kg bag (FCFA)
Urea	$CO(NH_2)_2$, assumed pure	18 000
Ammonium sulphate	$(NH_4)_2SO_4$, assumed pure	15 000
NPK 20–10–10	20 % N, 10 % P_2O_5, 10 % K_2O by mass	22 000

Data pack 3 – Rainfall and stream report. August rainfall was 310 mm, with three storms in the week after top-dressing. The stream draining the plain showed a dense green algal mat, a strong smell of decay, and dissolved oxygen below 2 mg L^{-1} at dawn (healthy streams usually hold above 6 mg L^{-1}). Dead tilapia were found along the banks.

The cooperative asks your group of student chemists to prepare a **one-page advisory note**: which fertilizer(s) to buy, at what cost, and how to prevent a repeat of the fish kill.

6.3 Tasks

Work through the following tasks in order. Show all working; give masses to 3 significant figures.

1. **Fertilizer analysis.** Using relative atomic masses (H = 1, C = 12, N = 14, O = 16, S = 32), calculate the percentage by mass of nitrogen in (a) urea $CO(NH_2)_2$ and (b) ammonium sulphate $(NH_4)_2SO_4$. Hence calculate, for each of the three fertilizers, the **cost per kilogram of nitrogen** delivered.

2. **Industrial chemistry.** Write balanced equations, with conditions, for the manufacture of:

   (a) ammonia by the Haber process from N_2 and H_2;

   (b) nitric acid from ammonia by the Ostwald process (three steps);

   (c) sulphuric acid by the Contact process (three steps, starting from sulphur);

   (d) urea from ammonia and carbon dioxide;

   (e) ammonium sulphate from ammonia and sulphuric acid.

State the oxidation number of nitrogen or sulphur in each reactant and product, and identify which steps are redox.

3. **The NPK question.** The NPK 20–10–10 supplies phosphorus as well as nitrogen. Given that the soil needs 80 kg N and 40 kg P_2O_5 per hectare, calculate the mass of NPK 20–10–10 needed per hectare to supply all the phosphorus, and the mass of nitrogen this provides. How much additional nitrogen must come from another fertilizer?

4. **Environmental chemistry.** Using the data pack, explain chemically the sequence of events from fertilizer application to fish deaths. Your answer must mention: dissolution and runoff of nitrate and ammonium ions, **eutrophication**, algal bloom, decomposition by aerobic bacteria, and the fall in dissolved oxygen. Include the equation for the oxidation of ammonium ions by nitrifying bacteria. Also explain, with an equation, why long-term use of ammonium sulphate can lower the soil pH, and why this matters for a soil already at pH 5.6.

5. **Recommendation.** Propose a costed fertilization plan for the whole 50 ha (choice of fertilizer(s), masses, number of bags, total cost in FCFA) and **two** pollution-control measures, each justified chemically or physically. Prepare your one-page advisory note and defend it in a 5-minute class debate.

6.4 Model answer

Task 1 – Percentage nitrogen and cost per kg N.

WORKED EXAMPLE 7 **Urea** $CO(NH_2)_2$

Molar mass $M = 12 + 16 + 2 \times (14 + 2 \times 1) = 60\ \text{g mol}^{-1}$. Mass of N per mole $= 2 \times 14 = 28$ g.

$$\%N = \frac{28}{60} \times 100 = 46.7\,\%.$$

A 50 kg bag contains $50 \times 0.467 = 23.3$ kg N. Cost per kg N $= 18\,000/23.3 = 772\ \text{FCFA kg}^{-1}$.

SOLUTION

Answer: 46.7% N; 772 FCFA kg^{-1} N.

WORKED EXAMPLE 8 **Ammonium sulphate** $(NH_4)_2SO_4$

Molar mass $M = 2 \times (14 + 4) + 32 + 4 \times 16 = 132\ \text{g mol}^{-1}$. Mass of N per mole $= 2 \times 14 = 28$ g.

$$\%N = \frac{28}{132} \times 100 = 21.2\,\%.$$

A 50 kg bag contains $50 \times 0.212 = 10.6$ kg N. Cost per kg N $= 15\,000/10.6 = 1\,415\ \text{FCFA kg}^{-1}$.

SOLUTION

Answer: 21.2% N; 1 415 FCFA kg^{-1} N.

WORKED EXAMPLE 9 **NPK 20–10–10**

Grade means 20% N by mass. A 50 kg bag contains 50 × 0.20 = 10.0 kg N. Cost per kg N = 22 000/10.0 = 2 200 FCFA kg^{-1} N.

SOLUTION

Answer: 20.0% N; 2 200 FCFA kg^{-1} N.

KEY POINTS **Cheapest nitrogen source**

Per kilogram of nitrogen, urea (772 FCFA) is by far the cheapest source, then ammonium sulphate (1 415 FCFA), then NPK (2 200 FCFA). However, only the NPK supplies the phosphorus the soil also needs.

Task 2 – Industrial equations with oxidation states and redox identification.

METHOD **Writing industrial equations for GCE**

1. Balance atoms and charge; add state symbols (g), (l), (aq), (s).
2. Write conditions above the arrow: catalyst, temperature, pressure.
3. Assign oxidation numbers to the key element (N or S) in every species.
4. A step is **redox** if oxidation numbers change.

Haber process (ammonia synthesis).

$$N_2(g) + 3H_2(g) \rightleftharpoons [Fe,\ 450\ ^{\circ}C,\ 200\ atm] 2NH_3(g) \qquad \Delta H = -92\ kJ\ mol^{-1}$$

Oxidation numbers: N_2: 0; H_2: 0; NH_3: N = −3, H = +1. **Redox**: Yes (N reduced 0 → −3; H oxidised 0 → +1).

Ostwald process (nitric acid from ammonia).

$$\text{Step 1: } 4NH_3(g) + 5O_2(g) \rightarrow [Pt/Rh,\ 900\ ^\circ C]4NO(g) + 6H_2O(g)$$
$$\text{Step 2: } 2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$
$$\text{Step 3: } 3\,NO_2(g) + H_2O(l) \longrightarrow 2\,HNO_3(aq) + NO(g)$$

Oxidation numbers of N: Step 1: NH_3 (-3) → NO (+2) oxidation; O_2 (0) → H_2O (-2) reduction. **Redox**. Step 2: NO (+2) → NO_2 (+4) oxidation; O_2 (0) → NO_2 (-2) reduction. **Redox**. Step 3: NO_2 (+4) → HNO_3 (+5) oxidation and → NO (+2) reduction (disproportionation). **Redox**.

Contact process (sulphuric acid from sulphur).

$$\text{Step 1: } S(s) + O_2(g) \longrightarrow SO_2(g)$$
$$\text{Step 2: } 2SO_2(g) + O_2(g) \rightleftharpoons [V_2O_5,\ 450\ ^\circ C]2SO_3(g) \qquad \Delta H < 0$$
$$\text{Step 3: } SO_3(g) + H_2SO_4(l) \rightarrow H_2S_2O_7(l) \xrightarrow{H2O} 2H_2SO_4(aq)$$

Oxidation numbers of S: Step 1: S (0) → SO_2 (+4) oxidation; O_2 (0) → SO_2 (-2) reduction. **Redox**. Step 2: SO_2 (+4) → SO_3 (+6) oxidation; O_2 (0) → SO_3 (-2) reduction. **Redox**. Step 3: S remains +6 throughout. **Not redox** (absorption/hydrolysis).

Urea manufacture.

$$2NH_3(g) + CO_2(g) \rightarrow [150 - 200\ ^\circ C,\ 150 - 250\ atm]CO(NH_2)_2(s) + H_2O(l)$$

Oxidation numbers: N in NH_3 and urea = -3; C in CO_2 and urea = +4. **Not redox**.

Ammonium sulphate manufacture.

$$2\,NH_3(g) + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

Oxidation numbers: N = -3 in both; S = +6 in both. **Not redox** (acid–base neutralisation).

WARNING **Common errors in industrial equations**

- Forgetting state symbols or conditions (catalyst, T, P) – these earn marks in GCE.
- Writing the Ostwald absorption as $NO_2 + H_2O \longrightarrow HNO_3$ (unbalanced). The correct stoichiometry is $3\,NO_2 + H_2O \longrightarrow 2\,HNO_3 + NO$.
- In the Contact process, writing $SO_3 + H_2O \longrightarrow H_2SO_4$ as the industrial step. Industrially, SO_3 is absorbed in concentrated H_2SO_4 to form oleum ($H_2S_2O_7$), then diluted.
- Confusing oxidation numbers: in NH_4^+, N is -3; in NO_3^-, N is +5.

Task 3 – The NPK calculation.

WORKED EXAMPLE 10 **Mass of NPK 20–10–10 required per hectare**

The soil needs 40 kg P_2O_5 per hectare. NPK 20–10–10 contains 10% P_2O_5.

$$m(\text{NPK}) = \frac{40\ \text{kg}}{0.10} = 400\ \text{kg ha}^{-1}.$$

This delivers 400×0.20 = 80 kg N per hectare – exactly the nitrogen requirement. Number of 50 kg bags per hectare = 400/50 = 8 bags.

SOLUTION

Answer: 400 kg ha^{-1} (8 bags/ha) supplies both N and P needs exactly. No additional nitrogen needed.

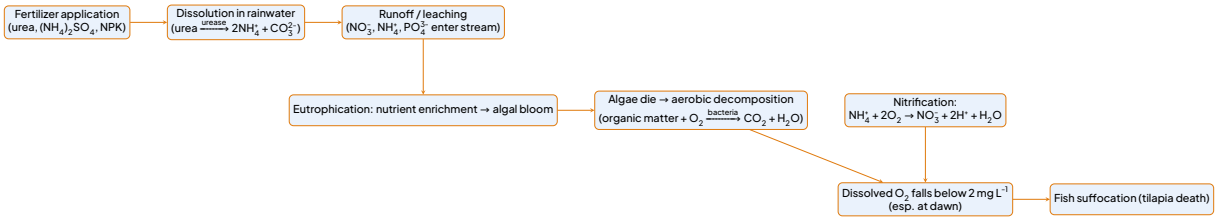
WORKED EXAMPLE 11 **Alternative blend: NPK + urea**

If cash is limited, supply part of the P from NPK and the rest of N from cheaper urea. Example: Use 200 kg NPK (4 bags) → 20 kg P_2O_5 and 40 kg N. Remaining N needed = 40 kg. Urea (46.7% N) required = 40/0.467 = 85.7 kg (≈ 1.7 bags). Cost per hectare: 4 × 22 000 + 1.7 × 18 000 = 118 600 FCFA (vs 176 000 FCFA for NPK alone).

SOLUTION

Note: The blend is cheaper but requires handling two products and still leaves a P deficit (only 20 of 40 kg P_2O_5 supplied). For full P requirement, NPK alone is simpler.

Task 4 – From fertilizer to fish deaths (eutrophication).



Sequence from fertilizer application to fish kill (eutrophication).

Chemical explanation:

1. **Dissolution and runoff:** Urea hydrolyses in soil (enzyme urease): $CO(NH_2)_2 + 2\,H_2O \longrightarrow 2\,NH_4^+ + CO_3^{2-}$. Ammonium sulphate dissociates: $(NH_4)_2SO_4 \longrightarrow 2\,NH_4^+ + SO_4^{2-}$. Nitrate ($NO_3^-$) from nitrification or NPK is highly soluble and not retained by soil. Heavy rains (310 mm) cause runoff/leaching, carrying NO_3^-, NH_4^+, and PO_4^{3-} (from NPK) into the stream.
2. **Eutrophication and algal bloom:** The sudden influx of N and P (limiting nutrients) causes ex-

plosive growth of algae – the observed green mat.

3. **Decomposition and oxygen depletion:** When algae die, aerobic bacteria decompose them: $C_xH_yO_z + O_2 \xrightarrow{\text{bacteria}} CO_2 + H_2O$, consuming dissolved oxygen.
4. **Nitrification in the stream:** Nitrifying bacteria (Nitrosomonas, Nitrobacter) oxidise ammonium, further consuming oxygen:

$$NH_4^+(aq) + 2\,O_2(g) \longrightarrow NO_3^-(aq) + 2\,H^+(aq) + H_2O(l)$$

5. **Fish kill:** Dissolved oxygen drops below 2 mg L^{-1} at dawn (respiration continues at night, no photosynthesis). Tilapia suffocate.

Soil acidification by ammonium sulphate: The nitrification of ammonium ions in soil releases H^+:

$$NH_4^+ + 2\,O_2 \longrightarrow NO_3^- + 2\,H^+ + H_2O$$

Additionally, sulphate (SO_4^{2-}) is not taken up by rice roots in significant amounts, so the accompanying acidity remains in the soil. On a soil already at pH 5.6 (slightly acidic), continued use of ammonium sulphate would lower the pH further, reducing availability of P, Mo, and beneficial microbial activity, ultimately depressing rice yields. Urea or NPK (which contains nitrate and ammonium but less acidifying potential per kg N) are safer choices here.

KEY POINTS **Eutrophication key points**

- Cause: Excess N and P (nutrient enrichment) from fertilizer runoff.
- Effect: Algal bloom → death → aerobic decomposition → oxygen depletion (hypoxia/anoxia).
- Fish die from lack of oxygen, not direct nitrate toxicity.
- Nitrification ($NH_4^+ \longrightarrow NO_3^-$) consumes O_2 and produces H^+, acidifying water and soil.

Task 5 – Recommendation (advisory note for the cooperative).

KEY POINTS **Recommended plan: NPK 20–10–10 only**

- Fertilizer: NPK 20–10–10.
- Rate: 400 kg ha^{-1} (8 bags/ha).
- Total for 50 ha: 400 bags = 20 000 kg.
- Total cost: 400 × 22 000 FCFA = 8 800 000 FCFA.
- Justification: Meets both N (80 kg/ha) and P_2O_5 (40 kg/ha) requirements exactly with a single product; soil K is adequate so the 10% K_2O is harmless. Avoids acidifying am-

monium sulphate. Simpler logistics (one product).

KEY POINTS **Alternative blend (if budget forces)**

- 200 kg/ha NPK (4 bags) + 85.7 kg/ha urea (1.7 bags).
- Supplies 20 kg P_2O_5 (half requirement) and 80 kg N.
- Cost: ≈ 118 600 FCFA ha^{-1} → 5 930 000 FCFA total.
- Drawback: Only half the P requirement met; yields may still be P-limited. Two products to transport and apply.

PROPERTY **Two pollution-control measures (chemically justified)**

1. Split application and weather forecasting: Apply fertilizer in three splits (basal, early tillering, panicle initiation) and never within 48 h of forecast heavy rain. This reduces the pool of soluble NO_3^- and NH_4^+ present at any one time, minimising the mass available for runoff during storms.
2. Vegetated buffer strips and paddy bunding: Maintain an unfertilized grass strip ≥ 5 m wide along the stream, and keep paddy bunds intact. Plant roots and soil microbes in the buffer take up NO_3^- and NH_4^+ (phyto-remediation) and trap sediment carrying particulate P before water reaches the stream. Bunds increase residence time, promoting denitrification ($NO_3^- \longrightarrow N_2$) in anaerobic zones.

WARNING **Common errors in this activity**

- Comparing fertilizer prices per bag instead of **per kg of nutrient**.
- Confusing NPK grade: 20–10–10 means %N - $\%P_2O_5$ - $\%K_2O$, not elemental P and K.
- In the Ostwald process, writing the absorption step as $NO_2 + H_2O \longrightarrow HNO_3$ (unbalanced, misses NO recycle).
- Stating that eutrophication kills fish by nitrate toxicity. The cause is oxygen depletion from bacterial decomposition and nitrification.
- Forgetting that urea hydrolysis initially raises pH (produces OH^- via $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$), but subsequent nitrification acidifies.

DID YOU KNOW?

Ndop Plain and rice in Cameroon

The Ndop Plain (North West Region) is a volcanic floodplain with heavy clay soils (vertisols) ideal for irrigated rice. It produces over 40% of Cameroon's rice. Fertilizer use has intensified since the 2000s, but soil organic matter is declining and stream pollution is a growing concern. The Upper Noun Valley Development Authority (UNVDA) promotes integrated soil fertility management (ISFM) combining mineral fertilizer, organic matter, and water control – exactly the approach your advisory note should advocate.

7 METHODS AND WORKED EXAMPLES

7.1 Skill 1: Assigning Oxidation Numbers to Sulphur and Nitrogen

METHOD **Assigning oxidation numbers**

To determine the oxidation number (o.n.) of S or N in any species:

1. Recall the fixed values: **oxygen** is −2 (except in peroxides, −1), **hydrogen** is +1 (except in metal hydrides, −1), Group 1 metals +1, Group 2 metals +2.
2. Apply the rule: the **sum of the oxidation numbers** equals the overall charge of the species (0 for a neutral molecule).
3. Set up a simple equation with x = unknown oxidation number and solve.
4. Check that the answer lies within the expected range: S from −2 to +6; N from −3 to +5.

Reflex: for oxoanions such as SO_4^{2-} or NO_3^-, the sum must equal the charge of the ion, not zero.

WORKED EXAMPLE 12 **Oxidation numbers of S and N**

Determine the oxidation number of sulphur in H_2S, SO_2, SO_3, H_2SO_4 and $S_2O_3^{2-}$, and of nitrogen in NH_3, N_2, NO, NO_2 and NO_3^-.

SOLUTION

Apply the sum rule in each case, taking o.n.(O) = −2 and o.n.(H) = +1.

- H_2S: $2(+1) + x = 0 \Rightarrow x = -2$. Sulphur is in its lowest state, −2.
- SO_2: $x + 2(-2) = 0 \Rightarrow x = +4$.
- SO_3: $x + 3(-2) = 0 \Rightarrow x = +6$, the maximum for sulphur.
- H_2SO_4: $2(+1) + x + 4(-2) = 0 \Rightarrow x = +6$.
- $S_2O_3^{2-}$ (thiosulphate): $2x + 3(-2) = -2 \Rightarrow 2x = +4 \Rightarrow x = +2$ (average value).
- NH_3: $x + 3(+1) = 0 \Rightarrow x = -3$, the minimum for nitrogen.
- N_2: elemental form, $x = 0$.
- NO: $x + (-2) = 0 \Rightarrow x = +2$.
- NO_2: $x + 2(-2) = 0 \Rightarrow x = +4$.
- NO_3^-: $x + 3(-2) = -1 \Rightarrow x = +5$, the maximum for nitrogen.

Conclusion: sulphur shows the states −2, +2, +4, +6; nitrogen shows every integer state from −3 to +5. Species in intermediate states (SO_2, NO) can be both oxidised and reduced: they can act as reducing agents or as oxidising agents depending on the partner.

WARNING **Frequent traps with oxidation numbers**

- In $S_2O_3^{2-}$ the value +2 is an average: the two sulphur atoms are not equivalent (one is in state +5-like, the other −1-like in the structural picture), but at GCE level the average value obtained from the sum rule is what is required.
- Do not forget that the sum equals the charge for ions: writing $x + 3(-2) = 0$ for NO_3^- instead of $= -1$ is the commonest error.
- An element in its maximum state (SO_3, H_2SO_4, HNO_3) can only be reduced: such species are oxidising agents only.

7.2 Skill 2: Writing and Balancing Redox Half-Equations

METHOD **Balancing a redox equation by half-equations**

1. Identify the species oxidised (o.n. increases) and the species reduced (o.n. decreases).
2. Balance the atoms of the element changing oxidation state.
3. Balance O by adding H_2O, then H by adding H^+ (acidic medium).
4. Balance the charge by adding electrons e^-.
5. Multiply the half-equations so that the electrons cancel, then add and simplify.
6. Check: atoms and total charge must balance on both sides.

Reflex: SO_2 is usually a reducing agent (S: +4 → +6); concentrated H_2SO_4 and HNO_3 are oxidising agents.

WORKED EXAMPLE 13 **Oxidation of sulphur dioxide by acidified dichromate**

In the laboratory, $SO_2(g)$ decolourises acidified potassium dichromate solution, which turns from orange to green. Write the two half-equations and the overall ionic equation. ($Cr_2O_7^{2-}$ is reduced to Cr^{3+}; SO_2 is oxidised to SO_4^{2-}.)

SOLUTION

Reduction half-equation (dichromate to chromium(III), Cr: +6 → +3, gain of $3e^-$ per Cr, i.e. $6e^-$ for $Cr_2O_7^{2-}$):

$$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$$

Check: Cr balanced (2 each side); O balanced (7 in $Cr_2O_7^{2-}$, 7 in H_2O); charge: left −2 + 14 − 6 = +6; right 2 × (+3) = +6. ✓

Oxidation half-equation (sulphur dioxide to sulphate, S: +4 → +6, loss of $2e^-$):

$$SO_2 + 2\,H_2O \longrightarrow SO_4^{2-} + 4\,H^+ + 2\,e^-$$

Check: S balanced; O: 2 + 2 = 4 each side; H: 4 each side; charge: left 0; right −2 + 4 − 2 = 0. ✓

Combine: multiply the oxidation half-equation by 3 so that both involve $6e^-$:

$$3\,SO_2 + 6\,H_2O \longrightarrow 3\,SO_4^{2-} + 12\,H^+ + 6\,e^-$$

Add the two half-equations and cancel e^-, then simplify H^+ (14 − 12 = 2 on the left) and H_2O (6 − 7: one water remains on the right):

$$Cr_2O_7^{2-} + 3\,SO_2 + 2\,H^+ \longrightarrow 2\,Cr^{3+} + 3\,SO_4^{2-} + H_2O$$

Final check: atoms balanced (Cr 2, S 3, O 11, H 2 each side); charge: left −2 + 0 + 2 = 0; right 2(+3) + 3(−2) + 0 = 0. ✓

The colour change orange → green confirms that SO_2 acted as a **reducing agent**. This reaction is also a standard test for SO_2 in qualitative analysis.

7.3 Skill 3: Calculations on the Contact Process

METHOD **Yield and mass calculations in the Contact process**

1. Write the three key steps: burning of sulphur $S(s) + O_2(g) \longrightarrow SO_2(g)$; catalytic oxidation $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$ (V_2O_5 catalyst, about 450 °C, 1–2 atm); absorption of SO_3 in concentrated acid $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$ (oleum), then dilution $H_2S_2O_7 + H_2O \longrightarrow 2\,H_2SO_4$.
2. Overall: 1 mol S gives 1 mol H_2SO_4 (sulphur is conserved through the chain).
3. Use $n = \frac{m}{M}$, scale by the mole ratio, then apply the percentage yield:

$$\text{actual mass} = \text{theoretical mass} \times \frac{\%\ \text{yield}}{100}$$

4. Quote the conditions and the reason for them (compromise between rate and equi-

librium position).

WARNING **Why SO_3 is not dissolved directly in water**

SO_3 reacts with water extremely violently, producing a corrosive mist of H_2SO_4 droplets that is very difficult to condense. Industrially, SO_3 is therefore absorbed in concentrated sulphuric acid to form oleum $H_2S_2O_7$, which is then diluted in a controlled way. Quoting this detail earns marks in "describe the Contact process" questions.

WORKED EXAMPLE 14 **Mass of sulphuric acid from sulphur**

A plant burns 64 tonnes of sulphur. Assuming the conversion to SO_3 is 98% efficient and absorption is complete, calculate the mass of pure sulphuric acid produced. (S = 32, O = 16, H = 1.)

SOLUTION

Step 1 — Moles of sulphur:

$$n(\mathrm{S}) = \frac{64 \times 10^6\ \mathrm{g}}{32\ \mathrm{g\,mol^{-1}}} = 2.0 \times 10^6\ \mathrm{mol}$$

Step 2 — Mole ratio: from the chain $S \longrightarrow SO_2 \longrightarrow SO_3 \longrightarrow H_2SO_4$, one mole of S gives one mole of H_2SO_4:

$$n(\mathrm{H_2SO_4})_{\mathrm{theoretical}} = 2.0 \times 10^6\ \mathrm{mol}$$

Step 3 — Apply the 98% conversion:

$$n(\mathrm{H_2SO_4})_{\mathrm{actual}} = 0.98 \times 2.0 \times 10^6 = 1.96 \times 10^6\ \mathrm{mol}$$

Step 4 — Convert to mass: $M(H_2SO_4) = 2(1) + 32 + 4(16) = 98\ \mathrm{g\,mol^{-1}}$

$$m = 1.96 \times 10^6 \times 98 = 1.9208 \times 10^8\ \mathrm{g} \approx 192\ \text{tonnes}$$

Answer: about 192 tonnes of pure H_2SO_4.
Remark: the oxidation of SO_2 is exothermic; a moderate temperature (~ 450 °C) is a **compromise**: lower temperatures favour the equilibrium yield but the reaction would be too slow even with the V_2O_5 catalyst. The heat released is recovered to maintain the catalyst beds at working temperature.

7.4 Skill 4: Explaining Acid Rain: Equations and pH

METHOD **Writing the chemistry of sulphur pollution**

1. Identify the source: burning of fossil fuels containing sulphur impurities, roasting of sulphide ores, volcanic emissions.
2. Write the sequence of atmospheric reactions:

$$S + O_2 \longrightarrow SO_2 \qquad 2\,SO_2 + O_2 \longrightarrow 2\,SO_3 \text{ (slow, catalysed by dust and } NO_x)$$

$$SO_3 + H_2O \longrightarrow H_2SO_4 \qquad SO_2 + H_2O \rightleftharpoons H_2SO_3$$

3. Conclude: rainwater contains H_2SO_4 and H_2SO_3, so its pH falls below 5 (unpolluted rain has pH ≈ 5.6 because of dissolved CO_2 forming weak carbonic acid).
4. Link to effects: corrosion of limestone/marble ($CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + CO_2$), acidification of lakes and soils, leaching of toxic Al^{3+} ions, damage to leaves and reduced crop yields.

Reflex: always mention both the dry pathway (direct deposition of SO_2 gas on surfaces and lungs) and the wet pathway (acid rain).

WORKED EXAMPLE 15 **Acid rain attacking a limestone building**

A limestone ($CaCO_3$) statue in an industrial town loses 0.50 mol of $CaCO_3$ per year through reaction with acid rain. (a) Write the equation for the reaction with sulphuric acid. (b) Calculate the mass of limestone eroded per year. (c) State two other environmental effects of SO_2 pollution. (Ca = 40, C = 12, O = 16.)

SOLUTION

(a) Sulphuric acid reacts with calcium carbonate:

$$CaCO_3(s) + H_2SO_4(aq) \longrightarrow CaSO_4(s) + H_2O(l) + CO_2(g)$$

(b) $M(CaCO_3) = 40 + 12 + 3(16) = 100\ \text{g mol}^{-1}$

$$m = n \times M = 0.50 \times 100 = 50\ \text{g}$$

The statue loses $\boxed{50\ \text{g}}$ of limestone per year. (c) Two further effects:

- **Acidification of lakes and rivers**: fish eggs fail to hatch and aquatic life dies when the pH falls below about 5.
- **Damage to vegetation**: SO_2 destroys chlorophyll, yellows leaves and reduces crop yields; soils are acidified and aluminium ions toxic to roots are released.

Control measures (to quote): flue-gas desulphurisation with limestone ($CaCO_3 + SO_2 \longrightarrow CaSO_3 + CO_2$), use of low-sulphur fuels, and catalytic converters on vehicles (which also reduce the NO_x that catalyses SO_2 oxidation).

7.5 Skill 5: Calculations on Ammonia and the Haber Process

METHOD **Equilibrium/yield reasoning for the Haber process**

For $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$, $\Delta H < 0$:

1. Predict the effect of a change using **Le Chatelier's principle**: high pressure favours the side with fewer gas moles (4 → 2); low temperature favours the exothermic (forward) direction.
2. Explain the industrial compromise: ~ 200 atm, ~ 450 °C, iron catalyst (to increase the rate), with continuous recycling of unreacted N_2 and H_2.
3. For calculations: use the mole ratio from the balanced equation, $n = m/M$ or $n = V/V_m$ for gases, then apply the percentage conversion.

WORKED EXAMPLE 16 **Ammonia production from hydrogen**

In a Haber plant, 600 kg of hydrogen react with excess nitrogen and the measured yield of ammonia is 30%. Calculate the mass of ammonia obtained. (N = 14, H = 1.)

SOLUTION

Step 1 — Moles of hydrogen:

$$n(H_2) = \frac{600 \times 10^3\ \text{g}}{2\ \text{g mol}^{-1}} = 3.0 \times 10^5\ \text{mol}$$

Step 2 — Mole ratio: from $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$, 3 mol H_2 give 2 mol NH_3:

$$n(NH_3)_{\text{theoretical}} = \frac{2}{3} \times 3.0 \times 10^5 = 2.0 \times 10^5\ \text{mol}$$

Step 3 — Apply the 30% yield:

$$n(NH_3)_{\text{actual}} = 0.30 \times 2.0 \times 10^5 = 6.0 \times 10^4\ \text{mol}$$

Step 4 — Mass: $M(NH_3) = 14 + 3(1) = 17\ \text{g mol}^{-1}$

$$m = 6.0 \times 10^4 \times 17 = 1.02 \times 10^6\ \text{g} \approx 1.0\ \text{tonne}$$

Answer: $\boxed{m(NH_3) \approx 1.0\ \text{tonne}}$.

Comment: the low single-pass yield is acceptable industrially because unreacted N_2 and H_2 are **recycled** continuously after the NH_3 is liquefied out, so the overall conversion approaches completion.

7.6 Skill 6: Reactions of Nitric Acid as an Oxidising Agent

METHOD **Predicting products of HNO_3 with metals**

1. Concentrated HNO_3 with a metal (e.g. Cu) gives the metal nitrate, **brown NO_2** gas and water.
2. Dilute HNO_3 gives the metal nitrate, **colourless NO** (which browns in air: $2\,NO + O_2 \longrightarrow 2\,NO_2$) and water.
3. Balance using half-equations: $NO_3^- + 2\,H^+ + e^- \longrightarrow NO_2 + H_2O$ (conc.) or $NO_3^- + 4\,H^+ + 3\,e^- \longrightarrow NO + 2\,H_2O$ (dil.).
4. Remember: HNO_3 never gives H_2 with metals (it is an oxidising acid); N goes from +5 to +4 (NO_2) or to +2 (NO).

WORKED EXAMPLE 17 **Copper and concentrated nitric acid**

A piece of copper foil is dropped into concentrated nitric acid. A brown gas is evolved and the solution turns blue-green. (a) Identify the gas and the blue-green species. (b) Write the balanced equation. (c) Calculate the volume of gas, measured at s.t.p., produced by 3.175 g of copper. (Cu = 63.5; molar volume at s.t.p. = 22.4 $dm^3\,mol^{-1}$.)

SOLUTION

(a) The brown gas is nitrogen dioxide NO_2; the blue-green solution contains copper(II) ions Cu^{2+} (as $Cu(NO_3)_2$). (b) Half-equations:

$$Cu \longrightarrow Cu^{2+} + 2\,e^- \qquad NO_3^- + 2\,H^+ + e^- \longrightarrow NO_2 + H_2O$$

Double the second half-equation so that the electrons cancel, then add:

$$Cu(s) + 4\,HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2\,NO_2(g) + 2\,H_2O(l)$$

(c) Moles of copper:

$$n(Cu) = \frac{3.175}{63.5} = 0.0500\ mol$$

From the equation, $n(NO_2) = 2\,n(Cu) = 0.100$ mol.

$$V = n \times V_m = 0.100 \times 22.4 = 2.24\ dm^3$$

Answer: $\boxed{V(NO_2) = 2.24\ dm^3}$ at s.t.p.

Safety note: NO_2 is toxic and contributes to photochemical smog and acid rain ($4\,NO_2 + 2\,H_2O + O_2 \longrightarrow 4\,HNO_3$); the experiment must be carried out in a fume cupboard.

7.7 Skill 7: Percentage of Nutrient in a Fertiliser

METHOD **Computing %N, %P2O5 or %K2O in a fertiliser**

1. Write the formula and compute its molar mass M.
2. Count the mass of the nutrient element in one mole of the compound.
3. Apply: $\%\text{nutrient} = \dfrac{\text{mass of nutrient in 1 mol}}{M} \times 100$.
4. For mixtures (NPK fertilisers), multiply by the mass fraction of each component.

Reflex: the quality of a nitrogen fertiliser increases with its %N: compare NH_4NO_3 (35%), $(NH_4)_2SO_4$ (21%), $CO(NH_2)_2$ (urea, about 47%).

WORKED EXAMPLE 18 **Comparing two nitrogen fertilisers**

A farmer near Bafoussam can buy either ammonium nitrate NH_4NO_3 or ammonium sulphate $(NH_4)_2SO_4$ at the same price per bag. (a) Calculate the percentage by mass of nitrogen in each fertiliser. (b) Advise the farmer, stating one environmental risk of excessive fertiliser use. (N = 14, H = 1, O = 16, S = 32.)

SOLUTION

(a) Ammonium nitrate: $M(NH_4NO_3) = 14 + 4(1) + 14 + 3(16) = 80\ \text{g mol}^{-1}$. It contains 2 N atoms, i.e. 28 g of N per mole:

$$\%N = \frac{28}{80} \times 100 = 35\%$$

Ammonium sulphate: $M((NH_4)_2SO_4) = 2(14 + 4) + 32 + 4(16) = 132\ \text{g mol}^{-1}$. It contains 2 N atoms (28 g):

$$\%N = \frac{28}{132} \times 100 \approx 21.2\%$$

(b) At equal price, **ammonium nitrate** delivers more nitrogen per bag (35% against 21.2%), so it is the better buy for nitrogen supply. (Ammonium sulphate may still be preferred where the soil also lacks sulphur.)
Environmental risk: excess nitrate is washed by rain into rivers and lakes (**leaching**), causing **eutrophication**: algal blooms develop, and when the algae die and decay the bacteria consume the dissolved oxygen, fish die, and drinking water may exceed safe nitrate limits. Fertilisers should therefore be applied in measured doses, away from streams, and not just before heavy rain.

7.8 Skill 8: Back-Titration of Ammonium in a Fertiliser

METHOD **Back-titration of an ammonium salt**

1. Boil a known mass of fertiliser with a known excess of NaOH: $NH_4^+ + OH^- \longrightarrow NH_3 + H_2O$ (all the NH_3 is driven off).
2. Titrate the remaining OH^- with standard HCl: $H^+ + OH^- \longrightarrow H_2O$.
3. Compute: $n(NH_4^+) = n(NaOH)_{added} - n(NaOH)_{excess}$, where the excess is found from the titre (do not forget the dilution factor if the solution was made up to a volumetric flask and only a portion titrated).
4. Convert to moles of salt (mind the stoichiometry: $2\,NH_4^+$ per $(NH_4)_2SO_4$), then to mass and to percentage purity.

Reflex: never titrate NH_4^+ directly with an indicator — it is too weak an acid; the back-titration via NH_3 release is the standard GCE method.

WORKED EXAMPLE 19 **Purity of an ammonium sulphate fertiliser**

1.32 g of an ammonium sulphate fertiliser was boiled with 50.0 cm^3 of 1.00 mol dm^{-3} NaOH (excess). After cooling, the mixture was made up to 250 cm^3 in a volumetric flask. 25.0 cm^3 portions of this solution required 31.00 cm^3 of 0.100 mol dm^{-3} HCl for neutralisation. Calculate the percentage by mass of $(NH_4)_2SO_4$ in the fertiliser. (N = 14, H = 1, O = 16, S = 32.)

SOLUTION

Step 1 — Moles of NaOH added:

$$n(NaOH)_{added} = 1.00 \times \frac{50.0}{1000} = 0.0500\ \text{mol}$$

Step 2 — Moles of excess NaOH in one 25.0 cm^3 portion: since $HCl + NaOH \longrightarrow NaCl + H_2O$ is 1:1,

$$n(NaOH)_{excess,\,portion} = n(HCl) = 0.100 \times \frac{31.00}{1000} = 3.100 \times 10^{-3}\ \text{mol}$$

Step 3 — Scale up to the full 250 cm^3 (factor $\frac{250}{25.0} = 10$):

$$n(NaOH)_{excess,\,total} = 3.100 \times 10^{-3} \times 10 = 0.03100\ \text{mol}$$

Step 4 — NaOH that reacted with NH_4^+:

$$n(NaOH)_{reacted} = 0.0500 - 0.03100 = 0.01900\ \text{mol}$$

From $NH_4^+ + OH^- \longrightarrow NH_3 + H_2O$ (1:1), $n(NH_4^+) = 0.01900$ mol. Step 5 — Moles and mass of $(NH_4)_2SO_4$: each formula unit contains $2\,NH_4^+$:

$$n((NH_4)_2SO_4) = \frac{0.01900}{2} = 9.500 \times 10^{-3}\ \text{mol}$$

$$M((NH_4)_2SO_4) = 132\ \text{g mol}^{-1}; \qquad m = 9.500 \times 10^{-3} \times 132 = 1.254\ \text{g}$$

Step 6 — Percentage purity:

$$\%\,(NH_4)_2SO_4 = \frac{1.254}{1.32} \times 100 = 95.0\%$$

Answer: the fertiliser is $\boxed{95.0\%}$ ammonium sulphate by mass.
Exam lesson: a computed percentage above 100% signals an error. The two commonest mistakes in this classic experiment are forgetting the ×10 dilution factor from 25.0 to 250 cm^3, and forgetting the factor 2 between NH_4^+ and $(NH_4)_2SO_4$. Always finish with a sanity check: the mass of pure salt found must not exceed the mass of sample taken.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 5 **Multiple Choice Questions**

1. In which of the following compounds does sulphur have an oxidation state of +4?

A. H_2S

B. SO_2

C. SO_3

D. H_2SO_4

2. The catalyst used in the Contact process for the manufacture of sulphuric acid is:

A. Iron

B. Vanadium(V) oxide

C. Platinum

D. Nickel

3. Which nitrogen oxide is a brown gas at room temperature?

A. N_2O

B. NO

C. NO_2

D. N_2O_4

4. The Haber process for ammonia synthesis is carried out at high pressure because:

A. It increases the rate of reaction only.

B. It decreases the activation energy.

C. The forward reaction proceeds with a decrease in the number of moles of gas.

D. The reaction is endothermic.

EXERCISE 6 **True or False with Justification**

State whether each statement is True or False. Justify your answer with a chemical equation or a brief explanation.

1. Sulphur dioxide (SO_2) is a basic oxide.
2. Concentrated sulphuric acid acts as a dehydrating agent on sugar ($C_{12}H_{22}O_{11}$) leaving a black mass of carbon.
3. Ammonia gas can be dried by passing it through concentrated sulphuric acid.
4. Eutrophication is caused by the depletion of oxygen in water bodies due to excessive growth of algae fed by nitrates and phosphates.

EXERCISE 7 **Definitions and Key Terms**

Define the following terms precisely:

1. Acid rain (state the typical pH range and main acidic components).
2. Catalyst (mention its effect on activation energy and equilibrium position).
3. Percentage yield (give the formula).
4. Le Chatelier's Principle.

EXERCISE 8 **Direct Calculations**

1. Calculate the percentage by mass of nitrogen in urea, $CO(NH_2)_2$. (Atomic masses: C=12, O=16, N=14, H=1).
2. 64 tonnes of sulphur are burned in air and the SO_2 converted to H_2SO_4 via the Contact process with an overall yield of 95%. Calculate the mass of sulphuric acid produced. (Molar masses: S=32, H_2SO_4=98).
3. Write the balanced chemical equation, with state symbols, for the reaction between ammonia and nitric acid to form ammonium nitrate.
4. Assign the oxidation number of nitrogen in NH_4^+, N_2O, NO_2^-, and NO_3^-.

8.2 I practise

EXERCISE 9 **The Contact Process – Industrial Manufacture of H_2SO_4**

1. Write the balanced equation for the combustion of sulphur in air.

2. Write the balanced equation for the catalytic oxidation of SO_2 to SO_3. State the catalyst used, the optimum temperature, and the pressure used industrially.
3. Explain, using Le Chatelier's principle, why:
   (a) A temperature of about 450 °C is used (compromise).
   (b) The pressure is kept near atmospheric (not very high).
   (c) Excess air (oxygen) is used.
4. SO_3 is not dissolved directly in water. What is it dissolved in? Write the equation for the subsequent reaction with water to form H_2SO_4.

EXERCISE 10 **Redox Chemistry of Sulphur and Nitrogen**

1. Assign the oxidation state of sulphur in: H_2S, S_8, SO_2, SO_3, H_2SO_4, $Na_2S_2O_3$.
2. Assign the oxidation state of nitrogen in: NH_3, N_2O, NO, NO_2, HNO_3, NH_4NO_3 (for both N atoms).
3. In the reaction: $2\,H_2S(g) + SO_2(g) \longrightarrow 3\,S(s) + 2\,H_2O(l)$, identify the oxidising agent and the reducing agent. Write the two half-equations.
4. SO_2 can act as both an oxidising and a reducing agent. Give one example of a reaction where it acts as a reducing agent and one where it acts as an oxidising agent (with balanced equations).

EXERCISE 11 **The Haber Process – Ammonia Synthesis**

The reaction is: $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g) \quad \Delta H = -92\ kJ\,mol^{-1}$.

1. Predict and explain the effect of increasing the pressure on: (a) the yield of ammonia, (b) the rate of reaction.
2. Predict and explain the effect of increasing the temperature on: (a) the yield of ammonia, (b) the rate of reaction.
3. State the role of the iron catalyst. Does it affect the yield?
4. Calculation: 28 g of nitrogen gas reacts with excess hydrogen. If the conversion of nitrogen is 30%, calculate the volume of ammonia gas produced at s.t.p. (Molar volume at s.t.p. = 22.4 L mol^{-1}).

EXERCISE 12 **Fertilizers – Composition and Economics**

1. Calculate the percentage by mass of nitrogen in:
   (a) Ammonium nitrate, NH_4NO_3
   (b) Ammonium sulphate, $(NH_4)_2SO_4$
   (c) Urea, $CO(NH_2)_2$
2. A farmer needs to supply 46 kg of nitrogen per hectare. Calculate the mass of each fertilizer required to supply this amount of nitrogen.
3. If the market prices per 50 kg bag are: NH_4NO_3 = 18,000 FCFA, $(NH_4)_2SO_4$ = 15,000 FCFA, Urea = 20,000 FCFA, determine which fertilizer is the most economical per kilogram of nitrogen supplied. Show your working.
4. State two environmental problems associated with the excessive use of nitrogenous fertilizers.

8.3 I prepare for the GCE

EXERCISE 13 **Environmental Chemistry: Sulphur Oxides and Acid Rain – 15 marks**

1. A roasting plant processing zinc sulphide ore releases sulphur dioxide into the atmosphere.
   (a) Write the equation for the roasting of zinc sulphide. [2 marks]
   (b) Describe, with balanced equations, how SO_2 in the atmosphere leads to the formation of acid rain. [4 marks]
   (c) State three harmful effects of acid rain on the environment. [3 marks]
2. Limestone ($CaCO_3$) slurry is used in flue gas desulphurisation (scrubbing) to remove SO_2.
   (a) Write the balanced equation for the reaction between SO_2 and $CaCO_3$ in the presence of water/oxygen to form gypsum ($CaSO_4 \cdot 2\,H_2O$). [3 marks]
   (b) Explain why this method is preferred over using sodium hydroxide solution. [2 marks]
3. State one major use of sulphuric acid in Cameroon's industry. [1 mark]

EXERCISE 14 **Laboratory Preparation and Properties of Ammonia – 12 marks**

1. Describe the laboratory preparation of a dry sample of ammonia gas. Your answer must include:
   (a) The reagents used (solid). [1 mark]
   (b) A labelled diagram of the apparatus setup (describe in words if drawing not possible). [3 marks]
   (c) The drying agent used and why it is suitable. [2 marks]
   (d) The method of collection and the reason for this choice. [2 marks]
2. Give one chemical test for ammonia gas and the expected observation. [1 mark]
3. Explain why concentrated sulphuric acid cannot be used to dry ammonia gas. Write an equation for the reaction that would occur. [2 marks]
4. Write the ionic equation for the reaction of aqueous ammonia with $Cu^{2+}(aq)$ to form the deep blue complex. [1 mark]

EXERCISE 15 **Data Analysis: Eutrophication and Oxygen Sag Curve – 13 marks**

A stream flows past a fertilized farmland. Water samples were taken at five points (A, B, C, D, E) downstream from the farm entry point. The concentrations of nitrate ions (NO_3^-) and dissolved oxygen (DO) were measured.

Sampling Point	$[NO_3^-]$ (mg L^{-1})	DO (mg L^{-1})
A (Upstream)	2	9.0
B	15	7.5
C	45	3.0
D	30	5.5
E (Far downstream)	10	8.0

1. On graph paper (or describe the axes and plots), plot the concentration of NO_3^- and DO against the sampling points (A to E) on the same axes (use two y-axes or separate graphs). [3 marks]
2. Identify the point where pollution from the farm most likely enters the stream. Justify your answer using the data. [2 marks]
3. Explain the trend in dissolved oxygen concentration from point A to point C in terms of eutrophication. Include the role of bacteria and the biochemical oxygen demand (BOD). [4 marks]

4. Explain the recovery of dissolved oxygen levels from point C to point E. [2 marks]
5. State two measures a farmer in the Cameroon highlands (e.g., around Mount Bamboutos) could take to reduce nitrate leaching into streams. [2 marks]

9 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple Choice Questions**

1. Answer: B (SO_2)
Oxygen has oxidation state −2. Let x be the oxidation state of S:

- H_2S: $x + 2(+1) = 0 \Rightarrow x = -2$
- SO_2: $x + 2(-2) = 0 \Rightarrow x = +4$ ✓
- SO_3: $x + 3(-2) = 0 \Rightarrow x = +6$
- H_2SO_4: $2(+1) + x + 4(-2) = 0 \Rightarrow x = +6$

2. Answer: B (Vanadium(V) oxide, V_2O_5)
In the Contact process, V_2O_5 catalyses $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$. Platinum works but is far more expensive and is easily poisoned by impurities such as arsenic compounds; iron and nickel are used in other processes (Haber, hydrogenation).
3. Answer: C (NO_2)
NO_2 is a pungent brown gas. N_2O and NO are colourless; N_2O_4 is colourless (the brown colour of the equilibrium mixture $2\,NO_2 \rightleftharpoons N_2O_4$ comes from NO_2).
4. Answer: C
$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$: 4 moles of gas → 2 moles of gas. By Le Chatelier's principle, increasing the pressure shifts the equilibrium towards the side with fewer moles of gas, i.e. towards NH_3, increasing the yield. (High pressure also increases the rate, but the thermodynamic reason for choosing it is the decrease in mole number; a catalyst, not pressure, affects activation energy; the reaction is exothermic.)

SOLUTION **Exercise 2 — True or False with Justification**

1. FALSE. SO_2 is an acidic oxide (an oxide of a non-metal). It dissolves in water to form an acidic solution of sulphurous acid:

$$SO_2(g) + H_2O(l) \longrightarrow H_2SO_3(aq)$$

It reacts with bases, e.g. $SO_2 + 2\,NaOH \longrightarrow Na_2SO_3 + H_2O$, which is the behaviour of an acid anhydride, not a basic oxide.
2. TRUE. Concentrated H_2SO_4 removes the elements of water (H and O in the ratio 2:1) from sucrose, leaving a black spongy mass of carbon:

$$C_{12}H_{22}O_{11}(s) \rightarrow [\text{conc. } H_2SO_4]\,12C(s) + 11H_2O(l)$$

This is the classic "black snake" / charring of sugar demonstration.
3. FALSE. Ammonia is a basic gas and reacts with the acid:

$$2\,NH_3(g) + H_2SO_4(l) \longrightarrow (NH_4)_2SO_4(s)$$

The gas would be absorbed, not dried. Ammonia is dried using a basic drying agent, quicklime (CaO) or soda lime.
4. TRUE. Nitrate- and phosphate-rich run-off stimulates excessive algal growth (algal bloom). When the algae die, aerobic bacteria decompose them and consume dissolved oxygen (high BOD). The resulting oxygen depletion (hypoxia) kills fish and other aquatic life — this sequence is eutrophication.

SOLUTION **Exercise 3 — Definitions and Key Terms**

1. Acid rain: rain (or any precipitation) with a pH below that of normal rain (≈ 5.6, due to dissolved CO_2), typically in the range pH 4–5 (sometimes lower). Its main acidic components are sulphuric acid (H_2SO_4, from oxidation of SO_2) and nitric acid (HNO_3, from oxidation of NO / NO_2).
2. Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative reaction pathway of lower activation energy, without itself being consumed (it is regenerated at the end). It speeds up both forward and reverse reactions equally, so it does not change the position of equilibrium — equilibrium is simply reached faster.
3. Percentage yield:

$$\text{Percentage yield} = \frac{\text{actual mass (or moles) of product obtained}}{\text{theoretical mass (or moles) of product}} \times 100\,\%$$

4. Le Chatelier's Principle: if a constraint (change of concentration, pressure or temperature) is applied to a system at dynamic equilibrium, the system adjusts itself so as to oppose (partially counteract) the imposed change and restore a new equilibrium.

SOLUTION **Exercise 4 — Direct Calculations**

1. Percentage of nitrogen in urea $CO(NH_2)_2$:
Molar mass: $M = 12 + 16 + 2(14 + 2 \times 1) = 60\ \text{g mol}^{-1}$. Mass of N per mole: $2 \times 14 = 28$ g.

$$\%N = \frac{28}{60} \times 100 = \boxed{46.7\,\%}$$

2. Mass of H_2SO_4 from 64 tonnes of S (yield 95%):
Overall stoichiometry: $S \longrightarrow SO_2 \longrightarrow SO_3 \longrightarrow H_2SO_4$, i.e. 1 mol S gives 1 mol H_2SO_4.

$$n(S) = \frac{64 \times 10^6\ \text{g}}{32\ \text{g mol}^{-1}} = 2.0 \times 10^6\ \text{mol}$$

$$m_{\text{theoretical}}(H_2SO_4) = 2.0 \times 10^6 \times 98 = 196 \times 10^6\ \text{g} = 196\ \text{tonnes}$$

$$m_{\text{actual}} = 0.95 \times 196 = \boxed{186.2\ \text{tonnes}}$$

3. Ammonia + nitric acid:

$$NH_3(g) + HNO_3(aq) \longrightarrow NH_4NO_3(aq)$$

(The product may equally be written as the solid $NH_4NO_3(s)$ after evaporation; the ionic form is $NH_4^+(aq) + NO_3^-(aq)$.)

4. Oxidation numbers of nitrogen:

- NH_4^+: $x + 4(+1) = +1 \Rightarrow x = \mathbf{-3}$
- N_2O: $2x + (-2) = 0 \Rightarrow x = \mathbf{+1}$
- NO_2^-: $x + 2(-2) = -1 \Rightarrow x = \mathbf{+3}$
- NO_3^-: $x + 3(-2) = -1 \Rightarrow x = \mathbf{+5}$

SOLUTION **Exercise 5 — The Contact Process**

1. Combustion of sulphur:

$$S(s) + O_2(g) \longrightarrow SO_2(g)$$

2. Catalytic oxidation:

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g) \qquad \Delta H = -196\ \text{kJ mol}^{-1}$$

Catalyst: vanadium(V) oxide, V_2O_5; optimum temperature: about 450 °C; pressure: about 1–2 atm (near atmospheric).

3. Le Chatelier analysis:

(a) The forward reaction is exothermic, so a low temperature favours a high yield of SO_3. However, at low temperature the rate is too slow (and the catalyst is inactive below about 400 °C). 450 °C is a compromise between a good yield and an economically acceptable rate.

(b) The forward reaction decreases the number of gas moles (3 → 2), so high pressure would favour SO_3. But the conversion at 450 °C and atmospheric pressure is already about 98%; the extra cost and safety risks of high-pressure plant are not justified by the tiny yield gain.

(c) Excess air increases the concentration of O_2; by Le Chatelier's principle the equilibrium shifts to the right, increasing the conversion of the more valuable reactant, SO_2.

4. SO_3 is dissolved in concentrated sulphuric acid to form oleum (direct dissolution in water would produce a corrosive, hard-to-condense acid mist because the reaction is too violent):

$$SO_3(g) + H_2SO_4(l) \longrightarrow H_2S_2O_7(l)$$

The oleum is then diluted:

$$H_2S_2O_7(l) + H_2O(l) \longrightarrow 2\,H_2SO_4(l)$$

SOLUTION **Exercise 6 — Redox Chemistry of Sulphur and Nitrogen**

1. Oxidation states of S: H_2S: −2; S_8: 0 (element); SO_2: +4; SO_3: +6; H_2SO_4: +6; $Na_2S_2O_3$: $2(+1) + 2x + 3(-2) = 0 \Rightarrow x = +2$ (average).
2. Oxidation states of N: NH_3: −3; N_2O: +1; NO: +2; NO_2: +4; HNO_3: +5; NH_4NO_3: −3 in NH_4^+ and +5 in NO_3^-.
3. $2\,H_2S(g) + SO_2(g) \longrightarrow 3\,S(s) + 2\,H_2O(l)$
S in H_2S goes from −2 to 0 (oxidation); S in SO_2 goes from +4 to 0 (reduction).
   - Oxidising agent: SO_2 (it is reduced).
   - Reducing agent: H_2S (it is oxidised).

Half-equations:

$$\text{Oxidation: } H_2S \longrightarrow S + 2\,H^+ + 2\,e^- \qquad \text{Reduction: } SO_2 + 4\,H^+ + 4\,e^- \longrightarrow S + 2\,H_2O$$

4. Dual behaviour of SO_2 (S in +4, an intermediate state):
   - As a reducing agent (S oxidised +4 → +6), e.g. with acidified potassium manganate(VII):

$$2\,KMnO_4 + 5\,SO_2 + 2\,H_2O \longrightarrow K_2SO_4 + 2\,MnSO_4 + 2\,H_2SO_4$$

     (purple solution decolourised), or simply $2\,SO_2 + O_2 \longrightarrow 2\,SO_3$.
   - As an oxidising agent (S reduced +4 → 0), e.g. with H_2S:

$$SO_2 + 2\,H_2S \longrightarrow 3\,S + 2\,H_2O$$

SOLUTION **Exercise 7 — The Haber Process**

1. Increasing pressure:

(a) Yield: increases. The forward reaction converts 4 moles of gas into 2 moles; by Le Chatelier's principle, compression shifts the equilibrium towards the side with fewer gas moles, i.e. towards NH_3.

(b) Rate: increases. Higher pressure means higher concentration of gas molecules, hence more frequent successful collisions per unit time.

2. Increasing temperature:

(a) Yield: decreases. The forward reaction is exothermic ($\Delta H < 0$); raising the temperature shifts the equilibrium in the endothermic (reverse) direction, decomposing NH_3.

(b) Rate: increases. More molecules possess energy $\geq E_a$, so the frequency of successful collisions rises sharply (the rate constant grows exponentially with T). Hence the industrial compromise at about 450 °C.

3. Iron catalyst: it provides an alternative pathway of lower activation energy, so equilibrium is reached faster. It does not affect the yield (position of equilibrium), since it accelerates forward and reverse reactions equally.
4. Volume of NH_3 at s.t.p.:

$$n(N_2) = \frac{28}{28} = 1.0\ \text{mol}$$
$$n(N_2)\ \text{reacted} = 0.30 \times 1.0 = 0.30\ \text{mol}$$
$$n(NH_3) = 2 \times 0.30 = 0.60\ \text{mol} \quad (\text{N2 : NH3 = 1:2})$$
$$V(NH_3) = 0.60 \times 22.4 = \boxed{13.44\ \text{dm}^3\ (\approx 13.4\ \text{dm}^3)}$$

SOLUTION **Exercise 8 — Fertilizers: Composition and Economics**

1. Percentage of nitrogen:

(a) NH_4NO_3: $M = 14 + 4 + 14 + 48 = 80$; mass of N = 28.

$$\%N = \frac{28}{80} \times 100 = 35.0\,\%$$

(b) $(NH_4)_2SO_4$: $M = 2(18) + 32 + 64 = 132$; mass of N = 28.

$$\%N = \frac{28}{132} \times 100 = 21.2\,\%$$

(c) $CO(NH_2)_2$: $M = 60$; mass of N = 28.

$$\%N = \frac{28}{60} \times 100 = 46.7\,\%$$

2. Mass of fertilizer to supply 46 kg of N:

$$m(NH_4NO_3) = \frac{46}{0.350} = 131.4\ \text{kg}$$
$$m((NH_4)_2SO_4) = \frac{46}{0.212} = 217.0\ \text{kg}$$
$$m(\text{urea}) = \frac{46}{0.467} = 98.5\ \text{kg}$$

3. Most economical fertilizer (cost per kg of N):
Cost per kg of fertilizer: NH_4NO_3: 18000/50 = 360 FCFA/kg; $(NH_4)_2SO_4$: 15000/50 = 300 FCFA/kg; urea: 20000/50 = 400 FCFA/kg.
Cost per kg of N supplied:

$$NH_4NO_3 : \frac{360}{0.350} = 1029\ \text{FCFA/kg N}$$
$$(NH_4)_2SO_4 : \frac{300}{0.212} = 1415\ \text{FCFA/kg N}$$
$$\text{Urea} : \frac{400}{0.467} = 857\ \text{FCFA/kg N}$$

Urea is the most economical per kilogram of nitrogen (857 FCFA/kg N), despite having the most expensive bag, because of its very high N content (46.7%).

4. Environmental problems of excess nitrogenous fertilizers:

- Eutrophication: nitrate leaching into rivers and lakes causes algal blooms, oxygen depletion and death of aquatic life.
- Groundwater contamination: high nitrate levels in drinking water (linked to methaemoglobinaemia in infants — "blue baby syndrome").
- (Also acceptable: soil acidification from ammonium-based fertilizers; release of the greenhouse gas N_2O.)

SOLUTION **Exercise 9 — Sulphur Oxides and Acid Rain (15 marks)**

1.(a) Roasting of zinc blende: [2]

$$2\,ZnS(s) + 3\,O_2(g) \longrightarrow 2\,ZnO(s) + 2\,SO_2(g)$$

[1 mark for correct formulae of reactants and products; 1 mark for balancing and state symbols.]

1.(b) Formation of acid rain: [4] SO_2 is first oxidised in the atmosphere (by O_2, catalysed by sunlight, dust or NO_2):

$$2\,SO_2(g) + O_2(g) \longrightarrow 2\,SO_3(g)$$

The SO_3 then dissolves in rainwater:

$$SO_3(g) + H_2O(l) \longrightarrow H_2SO_4(aq)$$

(Alternative route also credited: $SO_2 + H_2O \longrightarrow H_2SO_3$, then $2\,H_2SO_3 + O_2 \longrightarrow 2\,H_2SO_4$.) The rain therefore contains dilute sulphuric acid and falls with pH 4–5. [1 mark per correct balanced equation (2); 1 mark for mentioning atmospheric oxidation conditions; 1 mark for linking to acidic rain / low pH.]

1.(c) Three harmful effects of acid rain: [3]

- Acidifies lakes and streams, killing fish and aquatic organisms.
- Leaches nutrients (and toxic Al^{3+}) from soils; damages leaves and forests, reducing crop yields.
- Corrodes limestone/marble buildings and metallic structures (roofs, bridges).

[1 mark each, any three valid effects.]

2.(a) Flue gas desulphurisation: [3]

$$2\,SO_2(g) + 2\,CaCO_3(s) + O_2(g) + 4\,H_2O(l) \longrightarrow 2\,CaSO_4\cdot 2\,H_2O(s) + 2\,CO_2(g)$$

[1 mark reactants, 1 mark products including gypsum formula, 1 mark balancing.]

2.(b) Limestone is preferred because it is very cheap and abundant (often quarried locally), whereas NaOH is expensive; in addition the product, gypsum, is a useful saleable solid (plaster, cement, wallboard) rather than a waste solution requiring disposal. [2]
3. Use of H_2SO_4 in Cameroon: manufacture of fertilizers (e.g. phosphate/ammonium sulphate fertilizers); also acceptable: car battery acid, metal processing, petroleum refining. [1]

SOLUTION **Exercise 10 — Laboratory Preparation of Ammonia (12 marks)**

1.(a) Reagents: a mixture of solid ammonium chloride (NH_4Cl) and solid calcium hydroxide (slaked lime, $Ca(OH)_2$). [1]

$$2\,NH_4Cl(s) + Ca(OH)_2(s) \longrightarrow CaCl_2(s) + 2\,NH_3(g) + 2\,H_2O(g)$$

1.(b) Apparatus (description): [3] The solid mixture is placed in a round-bottomed (or test) flask held by a clamp, tilted slightly downwards at the mouth so that condensed water does not run back onto the hot glass. A delivery tube leads the gas first through a drying tower/U-tube packed with quicklime, then into an inverted (upside-down) dry gas jar. Heat is applied gently with a Bunsen burner. [1 mark heat source + reactants in flask; 1 mark drying unit in line; 1 mark upward delivery / inverted collection jar.]
1.(c) Drying agent: quicklime, CaO (or soda lime). It is suitable because it is a basic drying agent and therefore does not react with the basic gas NH_3. [2]
1.(d) Collection: by upward delivery (downward displacement of air) in an inverted gas jar, because ammonia is less dense than air and is extremely soluble in water (so it cannot be collected over water). [2]
2. Test: NH_3 turns damp red litmus paper blue (it is the only common alkaline gas); alternatively, it gives dense white fumes of NH_4Cl with a rod dipped in concentrated HCl. [1]
3. Concentrated H_2SO_4 is acidic and reacts with the basic ammonia instead of drying it: [2]

$$2\,NH_3(g) + H_2SO_4(l) \longrightarrow (NH_4)_2SO_4(s)$$

4. Deep blue complex with Cu^{2+}: [1]

$$Cu^{2+}(aq) + 4\,NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq)$$

(Examiner's note: credit also the two-step answer showing the initial pale blue precipitate $Cu(OH)_2$ dissolving in excess ammonia to give the deep blue tetraamminecopper(II) complex.)

SOLUTION **Exercise 11 — Eutrophication and Oxygen Sag Curve (13 marks)**

1. Graph: [3] *x*-axis: sampling points A →

E (equally spaced). Left y-axis: $[NO_3^-]$ (mg L^{-1}), 0–50; right y-axis: DO (mg L^{-1}), 0–10. The nitrate curve rises steeply from A to a peak at C (45), then falls through D to E. The DO curve falls from 9.0 (A) to a minimum of 3.0 at C, then recovers through D to 8.0 at E — the classic oxygen sag curve, roughly a mirror image of the nitrate curve. [1 mark correct axes with labels/units; 1 mark correct nitrate plot; 1 mark correct DO plot.]

2. Entry point of pollution: [2] Between A and B (most evident at B). Justification: $[NO_3^-]$ jumps from 2 mg L^{-1} (A, upstream background) to 15 mg L^{-1} at B, and DO simultaneously begins to fall (9.0 → 7.5). The sharp rise in nitrate just downstream of the farm identifies the run-off entry point.

3. Explanation of the DO fall from A to C (eutrophication): [4]

- Nitrate leached from the fertilized field is a plant nutrient; it causes an algal bloom (rapid growth of algae and aquatic plants) near B–C.
- The dense surface growth also shades submerged plants, reducing photosynthetic oxygen production.
- When the algae die, they are decomposed by aerobic bacteria, whose respiration consumes large amounts of dissolved oxygen — the biochemical oxygen demand (BOD) rises sharply.
- Oxygen consumption by decomposition exceeds re-oxygenation from the air and photosynthesis, so DO falls to its minimum (3.0) at C; fish and other aerobic organisms suffocate.

[1 mark nutrient → algal bloom; 1 mark death of algae; 1 mark aerobic bacterial decomposition raising BOD; 1 mark consumption > supply ⇒ DO minimum.]

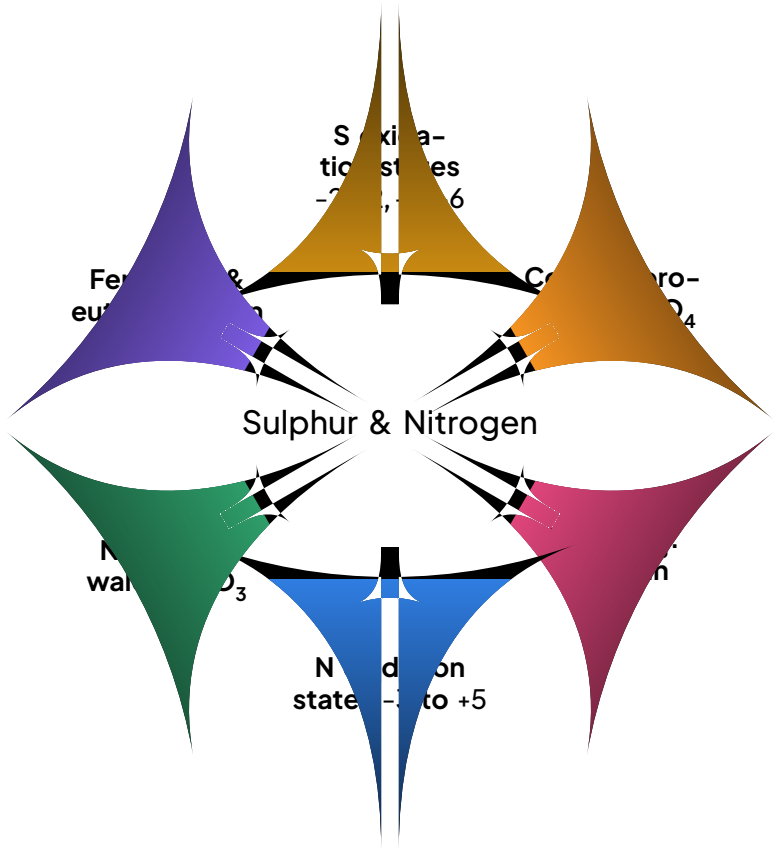
4. Recovery from C to E: [2] Downstream, the nitrate is diluted and taken up by plants, so algal growth and the load of dead organic matter decrease; bacterial activity and BOD fall. Meanwhile the stream is continuously re-aerated by dissolution of oxygen from the atmosphere (and photosynthesis resumes), so DO rises back towards its natural level (8.0 at E).

5. Two measures for a farmer around Mount Bamboutos: [2]

- Apply fertilizer in split doses matched to crop needs and avoid spreading just before heavy rain, so less nitrate is left to leach.
- Maintain buffer strips of grass/vegetation along streams and practise contour ploughing or terracing on the steep volcanic slopes to trap run-off.

(Also credited: using organic manure/compost instead of excess mineral fertilizer; planting cover crops in the off-season.)

10 SUMMARY SHEET



Mind map of the chapter: sulphur and nitrogen in the environment.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Oxidation state**: the charge an atom would have if all its bonds were fully ionic; sulphur shows -2 (H_2S), +2 (SO), +4 (SO_2, H_2SO_3), +6 (SO_3, H_2SO_4); nitrogen ranges from -3 (NH_3) to +5 (HNO_3).
- **Acid rain**: rain with pH below about 5.6, caused mainly by dissolved SO_2 and nitrogen oxides forming H_2SO_4 and HNO_3.
- **Eutrophication**: excessive enrichment of water with nitrates and phosphates, causing algal blooms, oxygen depletion and death of aquatic life.
- **Catalyst**: a substance that increases reaction rate without being consumed (V_2O_5 in the Contact process, Fe in the Haber process, Pt/Rh in the Ostwald process).

Key reactions and processes to know

- Contact process: $S(s) + O_2(g) \longrightarrow SO_2(g)$; $2SO_2(g) + O_2(g) \rightleftharpoons [V_2O_5, 450^\circ C] 2SO_3(g)$ (exothermic, moderate temperature, high pressure not needed); $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$ (oleum), then $H_2S_2O_7 + H_2O \longrightarrow 2\,H_2SO_4$. Never add SO_3 directly to water (acid mist).
- Properties of concentrated H_2SO_4: dehydrating agent (chars sugar, dries gases), oxidising agent with hot metals: $Cu + 2\,H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2\,H_2O$; dilute acid behaves as a typical dibasic acid.
- Sulphur dioxide: $SO_2 + H_2O \rightleftharpoons H_2SO_3$; bleaches by reduction; reducing agent (decolourises acidified $KMnO_4$).
- Haber process: $N_2(g) + 3H_2(g) \rightleftharpoons [Fe, 450^\circ C, 200 atm] 2NH_3(g)$, $\Delta H < 0$; compromise conditions.
- Test for NH_3: turns damp red litmus blue; white fumes with concentrated HCl. Test for NH_4^+: warm with NaOH, ammonia evolved.
- Ostwald process: $4\,NH_3 + 5\,O_2 \xrightarrow{Pt} 4\,NO + 6\,H_2O$; $2\,NO + O_2 \longrightarrow 2\,NO_2$; $3\,NO_2 + H_2O \longrightarrow 2\,HNO_3 + NO$.
- Dilute HNO_3 with Cu: $3\,Cu + 8\,HNO_3 \longrightarrow 3\,Cu(NO_3)_2 + 2\,NO + 4\,H_2O$; concentrated acid gives NO_2 (brown fumes).
- Fertilizers: NH_4NO_3, $(NH_4)_2SO_4$, urea $CO(NH_2)_2$, NPK; percentage of nitrogen in $NH_4NO_3 = \frac{28}{80} \times 100 = 35\%$.

Methods

- Assign oxidation states from rules (O usually −2, H usually +1), then identify oxidation (increase) and reduction (decrease).
- For industrial processes, always state: equation, catalyst, temperature, pressure, and justify conditions with Le Chatelier's principle.
- Environmental questions: name the pollutant, its source, the chemical equation of its formation, its effect, and one control measure (e.g. flue-gas desulphurisation with $CaCO_3$).

Common mistakes

- Writing $SO_3 + H_2O \longrightarrow H_2SO_4$ as the industrial step: industrially SO_3 is absorbed in concentrated H_2SO_4, not water.
- Confusing the bleaching action of SO_2 (reduction) with that of Cl_2 (oxidation).
- Saying concentrated H_2SO_4 liberates H_2 with metals: it is an oxidising acid and gives SO_2.
- Forgetting state symbols and conditions in the Contact and Haber equations.

- Attributing acid rain only to SO_2: nitrogen oxides (NO, NO_2) also contribute.

GCE tips

- Learn the three industrial processes with exact conditions; they are asked almost every year.
- In essay questions on pollution, structure your answer: source → equation → effect → remedy.
- Show all steps in percentage-nitrogen calculations and give the answer to 3 significant figures.
- Link fertilizer use in Cameroonian agriculture (maize, cocoa) to both increased yield and the risk of eutrophication of streams and lakes.

Signed • OnBuch+
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